PIC Requirements Traceability Matrix

AlphaHelion Procurement Intelligence Cloud. Specs: Technical Document AHG/PRC/PIC/2026-10/02 (TD) and Product Blueprint AHG/PRC/PIC/2026-10/01 (BP). Code at commit `5a63ed3` (+1 modified doc), read on 29 Oct 2026.

How this was built. Every row was checked against the code (`backend/app/pic/**`, migrations 0001–0007, `backend/tests/pic/**`, `frontend/src/app/pic/**`, `.github/workflows/*`, `deploy/**`), plus read-only SELECTs on the local full build `pic_neon_build` (8 tenants, every dataset pack). The project's status documents were used only as pointers. Where they disagree with the code, the code wins. Example: `PIC_Remaining_Work_Report.md` lists punch-out, EDI, auctions and 4-way match as "not built", but they are built. The suite was not run. The last recorded full run (`scratchpad/final_suite.log`) shows **387 passed, 2 failed**; the code review report says those 2 tests were re-run and passed after their expectations changed.

Status key

- **DONE**: implemented, and covered by tests or seeded data.
- **PARTIAL**: something is missing; the row says what.
- **NOT DONE**: not built.
- **EXTERNAL**: the code or adapter is ready but needs a credential, a system, data or a decision from you.
- **N/A**: does not apply.

Abbreviations: `svc/` = `backend/app/pic/services/` , `plat/` = `backend/app/pic/platform/` , `api/` = `backend/app/pic/api/` , `tests/` = `backend/tests/pic/` , `fe/` = `frontend/src/` . DB = `pic_neon_build` .

Database facts used below:

- 115 tables. Row-level security is forced on 112 of them; the 3 exceptions are global tables (`tenant` , `process_definition` , `alembic_version`).
- 8 tenants: AXN, GMC and DBY (manufacturing), CON, ITS, LOG, HLT, GEN.
- 18 `dataset_load` rows.
- All 84 F1–F9 steps were executed, and all 13 gateways were taken both ways.
- 490 agent runs and 3,423 audit events.
- 229 integration messages, all POSTED.

Part 1 — Technical Document

TD §1.1 Engineering conventions

ID	Requirement	Status	Evidence / gap
TD-1.1a	TypeScript end-to-end (NestJS services, Next.js web); Python for ML	**PARTIAL**	Web: Next.js 16 / TS (`frontend/package.json`). Backend is a **Python 3.14 FastAPI modular monolith** (`backend/app/main.py`), not NestJS services. This is an architecture deviation (project DD-01) that the customer has not signed off.
TD-1.1b	UUIDv7 IDs; human-readable numbers per tenant and number range	**DONE**	`uuid.uuid7()` throughout; `plat/core.py:next_number/next_numbers` ; `number_range` has 224 rows (DB)
TD-1.1c	Money as integer minor units + ISO currency; FX rate and date stored with every converted amount	**PARTIAL**	Integer minor units: `*_minor` columns. `fx_rate` table holds 4 rates per tenant, all for one date (28 Oct 2026). **No document converts currency, and no rate or date is stored on any amount.** PO currency is hard-coded `"INR"` (`svc/order.py:_create_po`).
TD-1.1d	UTC storage; tenant time zone; factory calendar	**DONE**	`tenant.timezone` ; `plant.working_days/holidays` ; `clock.subtract_working_days` used by `svc/planning.py:_Reads.release_date`
TD-1.1e	Every write → audit event + outbox event in the same transaction	**DONE**	`plat/audit.py:record` , `plat/outbox.py:emit` (caller's session); `StateMachine.apply` audits every transition
TD-1.1f	Configuration over code: thresholds, tolerances, mandatory fields, approval chains as effective-dated rule data	**PARTIAL**	33 effective-dated, scoped rule keys (`seed/reference.py:RULE_SETS` , `plat/rules.py:resolve`). Mandatory fields are not configurable (no rule key). Hard-coded values remain: `purch_org="1000"` , `currency="INR"` , company code default `"1000"` , GST fallback 18% (7 occurrences in `svc/`).
TD-1.1g	AI doctrine: no agent commits spend outside its autonomy level and role	**DONE**	The only spend-committing agent (Touchless PO) is gated by `plat/agents.py:guard(L4, scope po:release, value ≤ ₹25 L)` at `svc/order.py:343` . All other agents recommend or draft.

TD §2.1 Technology stack

ID	Layer / choice	Status	Evidence / gap
TD-2.1a	Web: Next.js, React, TS, AlphaHelion tokens	**DONE**	`fe/app/pic/**` (45 pages); tokens `#0B1020` , `#1A56DB` in `fe/components/pic/*`
TD-2.1b	API: NestJS; GraphQL for read models, REST for commands	**PARTIAL**	REST only: 254 routes under `/v1` (FastAPI). **No GraphQL.**

ID	Layer / choice	Status	Evidence / gap
TD-2.1c	Workflow: Camunda 8 (BPMN/DMN)	**PARTIAL**	Replaced by a Postgres-native engine (`plat/workflow.py`: human tasks, timers, SLAs, escalation, versioned `process_definition`). No Camunda, no BPMN XML, no DMN files. Needs a decision: accept the replacement or adopt Camunda.
TD-2.1d	Data: PostgreSQL 16 (RLS), Redis, OpenSearch, ClickHouse	**PARTIAL**	PG 16 (`ci.yml` `postgres:16`) with forced RLS. **No Redis** (rate limits are in-memory per instance), **no OpenSearch** (SQL search), **no ClickHouse** (spend cube on Postgres).
TD-2.1e	Events: Kafka + outbox + schema registry	**PARTIAL**	Outbox with in-process consumers (`plat/outbox.py`). `_publish_external` is a stub that returns None. No Kafka, no Avro registry.
TD-2.1f	AI/ML: Python services (FastAPI), feature store, MLflow registry, LiteLLM gateway	**PARTIAL**	Models are in-process heuristics with version strings (`demand-es-v1`, `copilot-rules-v1`). **No feature store, no MLflow, and no LLM call anywhere.** `litellm==1.102.2` is in `requirements.txt` but never imported.
TD-2.1g	Integration: SAP (OData/IDoc), Oracle, Dynamics, Tally, Odoo; cXML/OCI; EDI; GST e-invoice	**PARTIAL**	See §12. SAP is a simulator only; the other ERP connectors need credentials.
TD-2.1h	Infra: GCP GKE, Cloud SQL/AlloyDB, Pub/Sub, Secret Manager, Cloud Armor; Terraform	**EXTERNAL**	`deploy/terraform` (asia-south1) and `deploy/k8s` exist but are not provisioned. The current target is Vercel + Neon Singapore (`backend/vercel.json`, region `sin1`). **Nothing is deployed** (docs show placeholder URLs).
TD-2.1i	Observability: OpenTelemetry, Grafana, structured audit stream	**PARTIAL**	OTel is optional (`PIC_OTEL_ENABLED`, `plat/observability.py`). `/metrics` with a ServiceMonitor. **No Grafana dashboards in the repo.** Audit stream = hash-chained `audit_event`.

TD §2.2 Architecture decisions

ID	Decision	Status	Evidence / gap
AD-01	Camunda 8 for approvals and long-running flows; process keys are part of the API	**PARTIAL**	Own engine in `plat/workflow.py` with process keys `pic.*` (`plat/process_defs.py`). Process keys are exposed: `GET /v1/process-definitions` and `/v1/documents/{type}/{ref}/process`. Camunda itself is not used.
AD-02	PostgreSQL RLS; every table has tenant_id; connection sets app.tenant_id; RLS tested in CI	**DONE**	Migration `0002_rls_roles_grants.py`. `db.py:100-101` runs `SET LOCAL ROLE pic_app` and `set_config('app.tenant_id')`. Tests: `tests/test_controls.py::test_rls_hides_other_tenant_rows` and `::test_rls_blocks_cross_tenant_writes`. CI runs them on every push. DB: RLS forced on 112/115 tables.
AD-03	Outbox + Kafka; idempotent consumers; versioned schemas in a registry	**PARTIAL**	Outbox plus `processed_event` (idempotent consumers), with retry back-off (`next_attempt_at`, migration 0007). **No Kafka; no registry**; `schema_version` is always 1.
AD-04	ClickHouse spend cube: < 3 s on 50 M lines; < 5 min CDC refresh	**NOT DONE**	The cube is a Postgres GROUP BY (`svc/savings.py:spend_cube`). SPD-01 `svc/spend.py:overview` **loads every spend line of 12 months into Python**, so it cannot meet 3 s at 50 M lines. Never measured: `tools/volume_gen.py --measure` exists but no run is recorded. Refresh runs on `invoice.posted`, which is near real time.

ID	Decision	Status	Evidence / gap
AD-05	MRP optional; in-platform engine or ERP planned-order import	**DONE**	`svc/planning.py:run` ; `svc/erp_import.py:import_planned_orders` (JSON and CSV). DB has `MR-1048` in mode `ERP_IMPORT` .
AD-06	LLM access only through the LiteLLM gateway	**NOT DONE**	No gateway client exists. `svc/copilot.py` only mentions `PIC_LLM_GATEWAY_URL` in its docstring and builds answers deterministically. Nothing calls a model vendor, so the constraint is not violated, but the gateway path is unbuilt.
AD-07	Effective-dated rule sets; documents store rule_set_version	**DONE**	`rule_set` (256 rows); `plat/rules.py:resolve` resolves most-specific first, deterministically. `rule_set_version` is stored on requisition, PO, invoice, contract, material_request, award, task and audit_event.

TD §3 Service decomposition

ID	Requirement	Status	Evidence / gap
TD-3a	One service per bounded context, each owning its data; cross-service reads only via events/read models	**PARTIAL**	Modules exist as `svc/*.py` inside one deployable. **Modules read each other's tables directly** (e.g. `svc/order.py` reads Supplier/Contract; `svc/savings.py:_spend` reads PoLine/PO). No import-linter rule enforces boundaries, despite project DD-01.
TD-3b	Owned entities per Table 3	**PARTIAL**	All listed entities have tables, except **legal_entity, purch_org and purch_group** (stored as strings) and **mrp_run_result** (simulation results are kept on `mrp_run`).
TD-3c	API surfaces and events per Table 3	**DONE**	See §10 and §11

TD §4 Multi-tenancy & organisation model

ID	Requirement	Status	Evidence / gap
TD-4a	Hierarchy tenant → legal entity/company code → plant → storage location; purch org/group; cost objects (CC, internal order, WBS, asset)	**PARTIAL**	Tables `company_code` , `plant` , `storage_location` , `cost_object` (type COST_CENTER, WBS, ORDER or ASSET). **No legal-entity, purchasing-org or purchasing-group master** (`purch_org` is hard-coded `"1000"`).
TD-4b	RLS tenant isolation; per-tenant KMS keys for field encryption	**PARTIAL**	RLS is DONE. Field encryption uses per-tenant keys derived by HKDF from one `PIC_FIELD_KEY` (`plat/crypto.py`). **Not KMS**, and there is no key version, so keys cannot be rotated.
TD-4c	Rules bind to scope (tenant, company code, plant, category, purchase type); most specific wins	**DONE**	`plat/rules.py:resolve` (specificity rank, then effective date, then created_at); duplicate versions refused
TD-4d	Industry template seed bundle at tenant creation, editable afterwards	**DONE**	`seed/templates.py` (6 packs, `merged_rules`), `GET /v1/admin/templates` . Five industry tenants were built from these templates (DB).

TD §5 Domain data model

ID	Requirement	Status	Evidence / gap
TD-5a	Core entities and key attributes (Table 4)	**DONE**	115 tables. Examples: supplier has `twin_ref`, `segment`, `risk_tier`, encrypted GSTIN; material has lot-size policy, safety stock and lead times; invoice has `irn` and `erp_doc_no`; audit has `on_behalf_of`, `prev_hash` and `hash`. The org gaps are listed under TD-4a.
TD-5.1	Explicit lineage `document_link` with typed link types	**DONE**	`document_link` (492 rows), `plat/core.py:link/link_many`, `GET /v1/lineage/{type}/{ref}`, Lineage tab (`fe/components/pic/object.tsx:LineageView`)
TD-5.2a	Free-text PR line ⇒ material_id IS NULL	**DONE**	DB CHECK `ck_pr_line_free_text_no_material` (0001)
TD-5.2b	Contract-priced PO line: price within tolerance or deviation_reason NOT NULL	**DONE**	Enforced in the service, not the DB: `svc/order.py:_create_po` and `_review_change` raise P07
TD-5.2c	GR received ≤ PO qty × (1 + over-delivery tolerance)	**DONE**	`svc/receiving.py:149-151, 217` (RCV-TOLERANCE); test `tests/test_operating_cases.py::test_short_shipment_inside_tolerance_completes_receipt` plus review tests
TD-5.2d	Invoice unique (tenant, cc, supplier, no., FY); fuzzy duplicate ±₹1 / ±3 days	**DONE**	`uq_invoice_tenant_id_company_code_supplier_id_supplier_invoice_no_fiscal_year`. DMN-03 `dupAmountMinor 100`, `dupDays 3`; `svc/payables.py:_duplicate_check`
TD-5.2e	Integration message unique (tenant, target_system, idempotency_key)	**DONE**	`uq_integration_message_tenant_id_target_system_idempotency_key`

TD §6 State machines (Table 5) and guards (Table 6)

ID	Object / transition	Status	Evidence / gap
SM-PR	Requisition Draft→Submitted→In Approval→Approved→Partially Ordered→Ordered→Closed, plus Rejected/Returned/Cancelled	**DONE**	`svc/requisition.py:SM` (also supports re-approval and cancel)
SM-PLO	Planned order Created→Firmed→Converted, or Deleted by the next run	**DONE**	`svc/planning.py` (firm l.370, convert l.399, delete l.256–262); DB has CREATED, FIRMED, CONVERTED and DELETED. Guards are written by hand, not with the `StateMachine` class.
SM-MR	Material request Draft→Duplicate Check→In Stewardship→Approved→Synced to ERP, or Rejected/Merged	**DONE**	`svc/mdm.py` (adds DUPLICATE_BLOCKED); DB: SYNCED_TO_ERP 18, MERGED 1, REJECTED 1
SM-SRC	Sourcing event Draft→Published→Open→Closed→Under Evaluation→Award Pending→Awarded, or Cancelled	**DONE**	`svc/sourcing.py:SM` (also new_round and reject_award)

ID	Object / transition	Status	Evidence / gap
SM-CON	Contract Draft→In Negotiation→In Approval→Out for Signature→Active→Expiring→Expired, or Terminated/Renewed	**DONE**	`svc/contract.py:SM` (also extend); DB has ACTIVE, EXPIRED, TERMINATED, DRAFT and IN_NEGOTIATION
SM-PO	PO Draft→In Approval→Released→Transmitted→Confirmed→Partially Received→Received→Invoiced→Closed, or Cancelled; any change creates a new version	**DONE**	`svc/order.py:SM`, `po_version`; DB CHECK `ck_purchase_order_status` matches TD §19.2
SM-GR	GR Posted→Quality Hold→Accepted/Partially Rejected/Rejected→Reversed	**DONE**	`svc/receiving.py:GR_SM`
SM-SES	SES Draft→Submitted→Approved/Rejected→Revoked	**DONE**	`svc/receiving.py:SES_SM`
SM-INV	Invoice Captured→Duplicate Check→Matching→Exception/Matched→Approved→Posted→Paid, or Rejected	**DONE**	`svc/payables.py:SM`
SM-SUP	Supplier per category Registered→Under Qualification→Qualified/Conditional→Preferred/Blocked/Exited	**DONE**	`svc/supplier.py:_set_category_status`, `supplier_category_status`, `POST /v1/suppliers/{code}/categories/{category}/status`. No PREFERRED rows in the seed.
SM-SAV	Savings Identified→Negotiated→Contracted→Realising→Validated, or Lost	**DONE**	`svc/savings.py:SM`
SM-409	Illegal transitions return 409	**DONE**	`plat/errors.py:illegal_transition`, `tests/test_controls.py`
G-1	Requisition Submitted→In Approval: budget check or soft-stop justification; **mandatory fields per rule set**; purpose code (P01)	**PARTIAL**	Budget (rule BUDGET) and purpose code are enforced. **Configurable mandatory fields do not exist**; there is no rule key for them.
G-2	Sourcing Draft→Published: invitees Qualified or Conditional-with-waiver; weights sum to 100, then locked	**DONE**	`svc/sourcing.py:publish` (l.137–149), `_require_eligible`
G-3	Sourcing Closed→Under Evaluation: close time reached; bids unsealed with audit `bids.unsealed`	**DONE**	`BIDS_UNSEALED` audit (l.331); early access is blocked with 423 and audited as `BID_ACCESS_DENIED` (l.237–244)
G-4	Contract Out for Signature→Active: all signatures; approvals done; price validities published	**DONE**	`svc/contract.py:activate` (l.324–335; signature document hash check)
G-5	PO Draft→Released: supplier not Blocked; approval or touchless; SoD for releaser	**DONE**	`PO_SUPPLIER_BLOCKED` (`svc/order.py:340`); approver chain excludes the creator (`initiator_id=po.created_by`, l.367)
G-6	PO Confirmed→new version: qty/price/date change → v n+1 and re-confirmation	**DONE**	`svc/order.py:change_order`, `po_version`, EDI ORDCHG
G-7	SES Submitted→Approved: approver ≠ PO creator ≠ supplier user	**DONE**	`svc/receiving.py:930-932`
G-8	Invoice Matched→Posted: no open exceptions; posting idempotency key reserved	**DONE**	`svc/payables.py:post`, `posting_key`; `tests/test_controls.py` exactly-once tests

ID	Object / transition	Status	Evidence / gap
G-9	Supplier Qualified→Conditional automatically when a mandatory certificate expires	**DONE**	`@on_timer("CERTIFICATE_EXPIRY")` in `svc/supplier.py:701–724`; D1 branch case

TD §7 Rules & decision tables

ID	Table	Status	Evidence / gap
DMN-01	Requisition approval route: < ₹25,000 catalogue/contract auto; ₹25k–2 L cost-object owner; 2–10 L + dept head; > 10 L + procurement head + finance controller; asset + capex committee	**DONE**	`seed/reference.py:RULE_SETS["DMN-01"]` (bands, `autoChannels`, `assetAdds`, `slaBusinessDays 2`). Open decision: band upper bounds are exclusive, so exactly ₹2 L or ₹10 L takes the larger chain (code review §7).
DMN-02	Touchless PO: contract/catalogue price, Qualified/Preferred, value below band, no split, late < 0.30, risk < 30	**DONE**	`RULE_SETS["DMN-02"]`, `plat/rules.py:touchless`, `svc/order.py:validate`. PO-77410: risk 18, touchless (DB). The "touchless band" value is not given by the spec; the default is **₹25 L** and needs your confirmation.
DMN-03	Invoice tolerance: price ±1% or ₹500 (lower); qty 0% (3-way); SES 0%; tax exact + IRN; duplicate	**DONE**	`RULE_SETS["DMN-03"]`; `svc/payables.py:match` (price cap on line value, tax rate + HSN + IRN l.428–441)
PO-RISK	Additive score 18/35/18/40/15/25/12, capped at 100, explainable	**DONE**	`RULE_SETS["PO-RISK"]` weights match the TD; `plat/rules.py:po_risk`; factors stored in `po.risk_factors`. Caveat: the late-delivery probability is read from the seeded twin snapshot, not a model (see §13).
DMN-ver	Decision tables versioned per tenant	**DONE**	Each tenant has its own `rule_set` rows (8 per key)

TD §8 Workflow implementation (BPMN F1–F9)

DB coverage: every one of the 84 steps was executed. All 13 gateways were taken both ways (`process_step` grouped by step_id and outcome). Instances by flow: F1 7, F2 56, F3 3, F4 17, F5 62, F6 58, F7 72, F8 32, F9 21.

ID	Flow / key	Status	Evidence / gap
F1	`pic.mrp-to-pr`: scheduled/planner start; Review exceptions (MRP_CONTROLLER); lot sizing; conversion with source determination	**DONE**	10 steps (`plat/process_defs.py`); F1-05 taken as live run, live import and simulation; `svc/planning.py`. Not scheduled: runs are started from the API only (no cron for "scheduled run").
F2	`pic.requisition`: DMN-01; SLA 2 business days; parallel MDM stewards	**DONE**	9 steps; F2-03 taken via contract, catalogue, material and free text; F2-05 yes and no
F3	`pic.sourcing-award`: close timer; award approval by value (CATEGORY_MGR, CPO, FINANCE)	**DONE**	`@on_timer("SOURCING_CLOSE")`; AWARD-APPROVAL bands at ₹50 L; F3-07 yes and no

ID	Flow / key	Status	Evidence / gap
F4	`pic.contract`: award or renewal (−120 d timer); Legal/Finance approval; signature; consumption alerts 80/90/100%	**DONE**	`CONTRACT_RENEWAL` / `CONTRACT_EXPIRY` timers (`svc/contract.py:488,515`); thresholds `[80,90,100]`. Signatures are internal; the e-sign provider is EXTERNAL (§12).
F5	`pic.po`: DMN-02 touchless; confirmation SLA timer	**DONE**	`@on_timer("PO_CONFIRMATION_SLA")` (24 h, PO-APPROVAL); F5-04 and F5-08 taken both ways
F6	`pic.receipt`: quality-managed flag; tolerance	**DONE**	F6-02 goods and service; F6-05 yes and no
F7	`pic.invoice`: DMN-03; resolve exception (AP_CLERK); **payment due timer**	**PARTIAL**	The flow is built (F7-04 taken both ways). **No payment-due timer**: payment runs start only by `POST /v1/payments/runs`.
F8	`pic.supplier-onboarding`: finance bank verify ×2; certificate expiry timers	**DONE**	`bankApprovers 2`; `CERTIFICATE_EXPIRY` timer; F8-06 taken as qualified, conditional and rejected
F9	`pic.material-request`: parallel stewards; duplicate gateway ≥ 0.92	**DONE**	MDM-DUP `blockAt 0.92`, `warnAt 0.75`, 4 domains; F9-04 taken as blocked, warning and override
F-ver	Definitions versioned; running instances finish on their version	**PARTIAL**	`process_instance.process_version` is stored, but only version 1 exists. Definitions are Python code (`plat/process_defs.py`), so a tenant cannot edit or version them.
F-audit	Every human-task completion audited with actor, decision, comment and evidence snapshot	**DONE**	`plat/workflow.py:complete` (decision APPROVE/REJECT/RETURN, `evidence`)
F-esc	Escalation: SLA breach → reminder → escalate to manager → reassign; configurable	**PARTIAL**	Rule ESCALATION: reminder at level 1, manager role added from level 2 (`@on_timer("TASK_SLA")`). **No reassignment**: the original approvers keep the task (open decision in the code review §7).

TD §9 Core algorithms

ID	Algorithm	Status	Evidence / gap
ALG-9.1	MRP netting: net = max(0, gross − supply + safety); lot size exact/fixed/EOQ/period/min-max with pack rounding; release = need − (PDT + GR days) on the factory calendar; PAST_DUE, BELOW_SAFETY, RESCHEDULE_IN/OUT; firmed orders never changed; simulation creates no documents; net-change	**PARTIAL**	`svc/planning.py:lot_size, net_material, run`. MR-1043 nets 12,000 − (1,400 + 1,200 + 600 + 0) + 200 = 9,000. CANCEL exception also built. **Buckets are daily only; the TD's "daily/weekly by MRP type" is not built.** Performance: see NFR-MRP.
ALG-9.2	Source determination: contract → quota → source list → info record → buyer queue; exclude Blocked / not Qualified	**DONE**	`svc/source.py:determine` (methods CONTRACT, QUOTA, SOURCE_LIST, INFO_RECORD, NONE; `ELIGIBLE` Qualified/Preferred)
ALG-9.3	TCO; weighted score with locked weights; award MIP with capacity from the twin, min share, max suppliers, **risk constraints**	**PARTIAL**	`svc/sourcing.py:_tco` (price + freight + duty + packaging + carrying + quality − payment-term value). `_optimise` is an exact enumeration, not a MIP solver, with capacity, min/max share and max suppliers. RFQ-2231 60/40 at weighted 400.8 is reproduced. **No risk constraints. One item per event** (`_require_one_line`).
ALG-9.4	Three-way match: qty vs accepted GR − invoiced; 2-way otherwise; price ≤ min(pct, abs); tax rate/HSN/IRN; exception with suggested action	**DONE**	`svc/payables.py:match` (also 4-way, SES and ERS). INV-TF-8812: quantity exception → credit note → posted, ERP document 5100004421 (DB).

ID	Algorithm	Status	Evidence / gap
ALG-9.5	Realised savings only from posted invoices; cost avoidance never in budget; validator ≠ owner (SOD-07)	**DONE**	Consumer `invoice.posted` → `savings-realised`; avoidance ledger entries have `counts=False`; `sod.guard("SAVINGS_VALIDATE")`

TD §10 API specification

ID	Convention / endpoint	Status	Evidence / gap
API-1	REST commands; GraphQL read models; JSON; OpenAPI 3.1 per service	**PARTIAL**	REST plus one FastAPI-generated OpenAPI 3.1 document (turned off in production by `PIC_API_DOCS`). **No GraphQL.**
API-2	Every POST accepts Idempotency-Key; replays within 24 h return the original response	**PARTIAL**	`plat/core.py:Idempotent` is wired into **26 of 138 POST routes** (`api/*: idempotent(...)`). The other 112 ignore the key, even though the web client sends one on every POST (`fe/lib/pic/api.ts:41`). **Records never expire**: there is no 24 h window and no purge.
API-3	ETag / If-Match; 412 on stale writes	**PARTIAL**	Only on purchase orders (`api/order_pay.py:73,107,122`, `api/order_ops.py:51`). No other document type has an ETag.
API-4	RFC 9457 problem+json with code, ruleRef, traceId	**DONE**	`plat/errors.py`, `plat/http.py`; `tests/test_production.py`
API-5	Cursor pagination, max page 200	**DONE**	`api/pagination.py` (`MAX_PAGE = 200`, opaque cursor)
API-6	OAuth 2.0 / OIDC bearer; scopes per module; tenant from the token, never the payload	**DONE**	`plat/auth.py` (PyJWT HS256 with scopes from roles), `plat/oidc.py` (OIDC code + PKCE). Tenant comes from token claims.
API-7	Table 8: all 32 core endpoints	**DONE**	All 32 exist, e.g. `POST /v1/mrp/runs`, `/planned-orders/{id}/convert`, `/requisitions/{id}/lines/{n}/promote-material`, `/sourcing/events/{id}/publish`, `/bids`, `/award`, `/contracts/{id}/activate`, `GET /contracts/{id}/consumption`, `/purchase-orders/{id}/validate`, `/confirmations`, `/versions`, `/asns`, `/goods-receipts`, `/service-entries/{id}/approve`, `/invoices/{id}/match`, `/post`, `GET /spend/cube`, `GET /savings`, `/copilot/ask`, `/agents/{agent}/runs`, `GET /audit/{type}/{id}`, `/integration/messages/{id}/replay`
API-10.2	Create PO from requisition returns inherited terms, validation (risk factors), lineage	**DONE**	`POST /v1/purchase-orders` (`api/order_pay.py`), `svc/order.py:validate`, `plat/core.py:upstream_chain`

ID	Convention / endpoint	Status	Evidence / gap
API-10.3	Invoice post is idempotent; a retry returns `replayed: true`	**DONE**	`POST /v1/invoices/{ref}/post` (`api/order_pay.py:414`), tests in `tests/test_controls.py`

TD §11 Event catalogue

ID	Requirement	Status	Evidence / gap
EV-1	All 23 catalogue events are emitted	**DONE**	`outbox.emit` covers every catalogue event (mrp.run.completed … savings.realised), plus 22 extra events (contract.amended, purchase_order.cancelled, lot.recalled, …)
EV-2	Envelope: tenantId, eventId, occurredAt, actor, objectRef, version, traceId	**DONE**	`plat/outbox.py:envelope`
EV-3	Kafka topics `pic.<tier>.<event>` ; Avro schemas, backward-compatible	**NOT DONE**	`_publish_external` stub; no registry
EV-4	Main consumers as in Table 9	**PARTIAL**	Only **10 of 23** catalogue events have outbox consumers: contract.activated, contract.consumption.threshold, goods_receipt.posted, invoice.exception.raised, invoice.posted (savings, spend, scorecard), purchase_order.changed, purchase_order.released (ERP posting, consumption), quality.usage_decision, sourcing.event.awarded, sourcing.event.published. **No ai-svc consumer exists at all.** requisition.approved → order/sourcing is a synchronous call (`planning.on_requisition_approved`). supplier.status.changed → order/sourcing is checked synchronously. purchase_order.released → spend-svc/portal is not consumed. audit-svc consumers are replaced by direct audit writes.

TD §12 Integration architecture

ID	Integration	Status	Evidence / gap
INT-SAP	SAP S/4HANA outbound PR/PO/GR/invoice/masters; inbound masters, stock, open POs, planned orders, payment status (OData, IDoc fallback)	**PARTIAL**	Only the **sandbox simulator** exists (`plat/connectors.py:SapS4` writes the `erp_document` table, with fault injection). **There is no real OData or IDoc client.** It must be built and verified against your SAP sandbox (EXTERNAL dependency).
INT-ORA	Oracle Fusion/EBS REST/FBDI	**EXTERNAL**	`OracleFusion` connector (6 message types, find-before-retry) is written but tested against mocks only. Needs `PIC_ERP_ORACLE_FUSION_*` credentials and mapping confirmation.
INT-D365	Dynamics 365 OData/Dataverse	**EXTERNAL**	`Dynamics365` connector (5 types), mocks only. Needs `PIC_ERP_DYNAMICS_365_*` .

ID	Integration	Status	Evidence / gap
INT-TALLY	Tally Prime XML/ODBC via on-prem agent	**EXTERNAL**	`TallyPrime` (XML import plus lookup), mocks only. Needs the Tally gateway and agent.
INT-ODOO	Odoo JSON-RPC	**EXTERNAL**	`Odoo` connector, mocks only. Needs `PIC_ERP_ODOO_*`.
INT-IN	Inbound ERP → platform (masters, stock, open POs, payment status)	**PARTIAL**	Planned-order import only (`svc/erp_import.py` , `POST /v1/integration/planned-orders/import[/csv]`). Payment status comes back only from the simulator. **No inbound master, stock or open-PO sync.**
INT-12.2	Exactly-once steps 1–4: message in the same transaction; skip if POSTED; query by external ref before retry; back-off, max 8, DEAD_LETTER, replay	**DONE**	`plat/integration.py` (2^n min back-off, `DEAD_LETTER` after 8 attempts, l.167–225), `POST /v1/integration/messages/{id}/replay` ; `tests/test_controls.py` , `test_review_platform.py` (F6)
INT-12.2-5	Nightly reconciliation per company code	**PARTIAL**	`svc/payables.py:reconcile` , `POST /v1/reconciliation/runs` , resolve endpoint; REC-101..103 in DB. **Not scheduled**: no nightly job. The Vercel cron runs only the worker tick, once a day (`backend/vercel.json` `0 3 * * *`).
INT-PUNCH	Punch-out cXML PunchOutSetupRequest/OrderMessage and SAP OCI; carts become catalogue PR lines	**EXTERNAL**	`svc/punchout.py` with a simulator; D7 laptop order ₹78,500 runs through it. Real supplier punch-out accounts are needed.
INT-EDI	EDI ORDERS/ORDCHG, ORDRSP, DESADV, **INVOIC** via gateway (AS2/SFTP)	**PARTIAL**	ORDERS/ORDCHG out, ORDRSP/DESADV in (`svc/transmission.py` , `POST /v1/edi/inbound` , module case PO-77425). **INVOIC inbound is not built.** Transport needs `PIC_EDI_GATEWAY_URL` (EXTERNAL).
INT-GST	GST e-invoice IRN and signed QR validated at capture; e-way bill on ASN	**EXTERNAL**	`svc/einvoice.py` : IRP/GSP adapter, plus local IRN hash and QR payload checks (the NIC signature is not verified locally). E-way bill rule ≥ ₹50,000 with format check (`svc/receiving.py:107`). Needs GSP credentials (`PIC_IRP_*`).
INT-RISK	Supplier risk feeds: sanctions, financial-risk, news	**PARTIAL**	Sanctions provider adapter (`PIC_SANCTIONS_*` , `svc/supplier.py:212-229`) with 4 fictitious SAMPLE-LIST names. **Financial and news feeds have only the `svc/risk.py:ingest` entry point; no provider adapter.**
INT-TWIN	Supplier Capability Digital Twin: read capability, evidence, free capacity, trust; write back OTIF/PPM/invoice accuracy	**PARTIAL**	Reads the local `twin_snapshot` (4 seeded rows) via `svc/supplier.py:twin` . Writes back to the local `twin_feedback` table (138 rows). **No connector to the actual Twin service** (no `PIC_TWIN_*` adapter).
MIG-1	Suppliers & materials: initial load + delta sync; counts, duplicate scan, bank-verification sample	**PARTIAL**	Legacy migration for GMC/DBY (`seed/legacy.py` , `tools/legacy_extract.py` , `tools/migration_report.py`). **No generic ERP master loader and no delta sync.** Real data is EXTERNAL.
MIG-2	Open contracts: template load + AI clause extraction with review; value/validity totals	**NOT DONE**	No contract loader and no clause extraction

ID	Integration	Status	Evidence / gap
MIG-3	Open POs with ERP doc no. as external ref; open value per supplier matches	**NOT DONE**	No open-PO loader (only legacy POs)
MIG-4	Spend history 24–36 months: bulk load + classification; ties to GL ±0.5%	**PARTIAL**	Classification runs on loaded history (AXN has 2,568 generated lines); `volume_gen` uses COPY. **No customer-file spend loader and no GL tie-out check.**
MIG-5	Open invoices as read-only references; AP ageing matches	**NOT DONE**	No loader

TD §13 AI & ML architecture

ID	Item	Status	Evidence / gap
ML-1	Spend classification: text classifier + embedding similarity	**PARTIAL**	Keyword/taxonomy rule classifier (`svc/spend.py:classify_text`) with confidence and steward queue below 0.80. **No ML model and no embeddings.** D9: accuracy 0.9936 overall; curated set 0.80; **auto-apply precision 0.857, below the proposed 0.95 gate (FAIL)**.
ML-2	Demand forecast: gradient boosted + seasonal baseline	**PARTIAL**	Exponential smoothing `demand-es-v1` (alpha 0.3, `svc/forecasting.py`). D9 MAPE h1 = 0.037 on 45 points. **No gradient-boosted model.**
ML-3	Material duplicate score: Siamese embedding + rules	**PARTIAL**	difflib plus Jaccard on description, combined with MPN, attribute and classification weights (`svc/mdm.py:32–35`). **No embedding model.** Supplier dedup precision 0.83 < 0.9 gate (FAIL).
ML-4	Should-cost: parametric models by commodity	**PARTIAL**	One parametric model, die-casting only (`svc/sourcing.py:should_cost`)
ML-5	Late-delivery probability classifier	**NOT DONE**	The value is read from `twin_snapshot.late_delivery_prob` (seeded, 4 rows); no model
ML-6	Supplier risk: scorecard + anomaly detection	**PARTIAL**	Threshold rules (SUPPLIER-RISK); **no anomaly detection**
ML-7	Invoice exception cause: rules + classifier	**PARTIAL**	Rules only (`svc/payables.py:match` , suggested actions)
ML-8	Contract redline: LLM with playbook retrieval	**PARTIAL**	Deterministic playbook comparison (CLM-PLAYBOOK positions STANDARD/FALLBACK/NON_STANDARD); **no LLM**
AGT-1	Agent registered with purpose, input events, tools/scopes, default autonomy, value/category limits, run-as role	**DONE**	`plat/agents.py:CATALOGUE` (15 agents × 8 tenants = 120 rows): value limit, categories, run_as_role. "Input events" are not registered.
AGT-2	Before any tool call, the guard checks role permission, **SoD**, autonomy, value limit and kill switch; denied calls become L2 drafts	**PARTIAL**	`agents.guard` checks kill switch, level, role scope, value, category and acceptance rate, **but not SoD** (SoD applies only through the underlying command's `sod.guard`). **It is called at only 2 sites** (Touchless PO, Spend Classification).
AGT-3	Every run stores inputs, evidence IDs, model/prompt version, output, confidence, action, feedback	**DONE**	`agent_run` (490 rows), `agents.log_run` , `POST /v1/recommendations/{id}/feedback`

ID	Item	Status	Evidence / gap
AGT-4	Kill switch per agent and per tenant; automatic suspension on low acceptance or drift	**PARTIAL**	Per-agent kill switch in each tenant (`PUT /v1/agents/{code}/kill-switch`). Suspension below min acceptance (70%) after ≥ 20 feedback events is built. **No tenant-wide switch and no drift detection** (the runbook `model-drift.md` exists but no code backs it).
CP-1	Copilot answer contract (answer, evidence, confidence, explanation, recommendedAction, permission)	**DONE**	`svc/copilot.py:_contract` ; D9 completeness 1.0 on 40 questions (gate 0.99)
CP-2	Retrieval tenant-scoped and permission-filtered before the model sees text	**DONE**	Runs inside an RLS transaction, filtered by scopes; supplier users see only their own documents
CP-3	PII and bank data redacted at the gateway; prompts/responses logged with per-tenant retention	**PARTIAL**	No gateway, so no redaction. Questions are logged as COPILOT agent runs (`api/insight.py:39`), but **no retention policy applies to them**.
CP-4	Numbers computed by tools, not by the model	**DONE**	SQL aggregates; there is no model

TD §14 Security & compliance

ID	Control	Status	Evidence / gap
SEC-ID	SSO via SAML/OIDC; MFA mandatory for approvers, finance and admins; supplier users invited with MFA	**PARTIAL**	OIDC is DONE (`plat/oidc.py`); **SAML is missing**. TOTP MFA is built (`plat/mfa.py`), but **rule MFA has `enforce=false` in all 8 tenants** (DB). Invited supplier users are not required to enrol MFA (`svc/invitations.py`).
SEC-AZ	RBAC + ABAC (company code, plant, category, value band) evaluated per request	**PARTIAL**	RBAC: 20 roles with scopes (`plat/auth.py`). ABAC is enforced **only on workflow tasks** (`plat/workflow.py:_can_work_as` : cost object, plant, category, steward domain). `auth.check_value_limit` and `check_attr` are defined but **never called**, so commands are not restricted by company code, plant or value band.
SEC-SOD	SoD SOD-01..07 checked at command time for humans and agents; conflicts block or require mitigating approval	**PARTIAL**	8 rules (SOD-01..07 plus 02a) guarded at command time (`plat/sod.py:guard`). Guarded actions: PO_CREATE, GR_POST, INVOICE_POST, PAYMENT_APPROVE, SOURCING_AWARD, MATERIAL_STEWARD_APPROVE, SAVINGS_VALIDATE, INVOICE_EXCEPTION_APPROVE. **MITIGATION mode just raises 409; there is no mitigating-approval path.**
SEC-DP	TLS 1.3; AES-256 at rest; field-level encryption with per-tenant keys; masked display	**PARTIAL**	Fernet plus HKDF per tenant (`plat/crypto.py`); GSTIN and bank data masked. TLS and at-rest encryption come from the host (EXTERNAL). **No KMS and no key versioning or rotation.**
SEC-BANK	Bank-change fraud: dual finance approval, call-back, cooling-off before first payment, alert on change + invoice within 30 days	**DONE**	`bankApprovers 2` ; `POST /v1/bank-changes/{n}/callback` ; `coolingOffHours 72` (`svc/supplier.py:524` , payment run skip `svc/payables.py:961`); `svc/risk.py:88` 30-day alert
SEC-AUD	Append-only, hash-chained audit: who/what/when/why, before/after, evidence, rule version, AI ref	**DONE**	`plat/audit.py` (advisory lock, sha256 chain), `REVOKE UPDATE, DELETE` (0002), `GET /v1/audit/verify` ; 3,423 events

ID	Control	Status	Evidence / gap
SEC-APP	OWASP ASVS L2; SAST/DAST/SCA in CI; dependency pinning; secrets in Secret Manager	**PARTIAL**	CodeQL, gitleaks, pip-audit gate and npm audit (`security.yml` , `ci.yml`). **ZAP is report-only** (`continue-on-error: true`). ASVS gaps per `PIC_Production_Readiness.md` : no breached-password check, no key rotation, no threat model. Frontend dependencies use `^` ranges (not pinned). Secret Manager via `deploy/k8s/external-secret.yaml` is EXTERNAL.
SEC-PRIV	DPDP: purpose limitation, India residency default, retention and deletion workflows	**PARTIAL**	Export, erasure (pseudonymisation) and a retention report (`svc/privacy.py`). **No retention purge job.** Retention periods are proposals that need legal review. Residency is currently Neon Singapore (decision needed).

TD §15 Non-functional requirements (target vs measured)

ID	NFR	Target	Measured	Status	Evidence / gap
NFR-AV	Availability	99.9% transactional, 99.5% analytics	—	**EXTERNAL**	Not deployed. K8s manifests have 2+ replicas, PDB and HPA; the real figure depends on the hosting plan.
NFR-P95	Transactional API latency	P95 < 400 ms	not measured	**NOT DONE**	k6 scripts in `tests/load/*.js` ; no run recorded (k6 binary downloaded, no results)
NFR-PR	PR form load	< 2 s	not measured	**NOT DONE**	No frontend performance test
NFR-CUBE	Spend cube on 50 M lines	< 3 s	not measured	**NOT DONE**	Postgres; SPD-01 loads every line into memory (`svc/spend.py:overview`). `volume_gen --measure` was never run at scale.
NFR-MRP	Net-change run, 50,000 materials	< 10 min (600 s)	**681 s** best (regenerative, 1 plant, local PG; `scratchpad/after_50k.log`); later reruns 1,105 s and 1,615 s; 5,000 materials in 179 s	**NOT DONE**	Target missed by 13% on the best run, and by 84–169% on reruns. `tools/perf_mrp.py` , migration 0006 indexes, bulk inserts.
NFR-SCALE	Concurrency and volume	10,000 concurrent users per tenant; 5 M invoices/year	not measured	**NOT DONE**	`tools/volume_gen.py` can generate 5 M invoices; no run recorded
NFR-SEC	SSO, MFA, RBAC+ABAC, field encryption, ASVS L2	—	—	**PARTIAL**	See SEC-ID, SEC-AZ, SEC-APP
NFR-TEN	Logical isolation with RLS; per-tenant keys	—	RLS forced on 112/115 tables	**DONE**	Per-tenant derived keys (not KMS; see SEC-DP)
NFR-AUD	Immutable audit log, 10-year retention configurable	10 y	default **8 y**	**PARTIAL**	Immutable is DONE. `svc/privacy.py` CATEGORIES default to 8 years; configurable via PRIVACY-RETENTION; **no archival or purge job**.

ID	NFR	Target	Measured	Status	Evidence / gap
NFR-RES	India region by default; tenant-level residency	India	Neon ap-southeast-1 (Singapore)	**EXTERNAL**	Decision needed. Terraform asia-south1 is ready. **Tenant-level residency (per-tenant DB or region) is not built.**
NFR-A11Y	Accessibility	WCAG 2.1 AA	not audited	**NOT DONE**	No axe/Lighthouse checks; only 15 aria/role attributes in the shared components
NFR-DR	Recovery	RPO 15 min, RTO 4 h	not tested	**EXTERNAL**	`docs/runbooks/backup-restore-dr.md`; needs Neon PITR plan and a DR drill
OBS-1	OTel traces web → API → services → workflow → connectors; traceId in every error	—	—	**PARTIAL**	traceparent propagated, and traceId is in every problem+json. OTLP export needs a collector (EXTERNAL). Connectors do not add outbound trace headers.
OBS-2	Golden signals per service; business SLOs (PR approval latency, touchless, match, posting)	—	—	**DONE**	`/metrics`, `pic_slo_*` gauges (`plat/observability.py:468-491`); `tests/test_production.py`
OBS-3	Runbooks: dead-lettered postings, workflow incidents, MRP failure, model drift	—	—	**DONE**	`docs/runbooks/` (10 files)
OBS-4	Blue/green deploys; migrations backward-compatible for one release	—	—	**PARTIAL**	K8s `rollingUpdate` (not blue/green); Vercel deploys are atomic. No automated check that migrations are backward-compatible.

TD §16 Front-end implementation

ID	Item	Status	Evidence / gap
FE-1	App Router workspace routes per module; screen IDs map to route components	**PARTIAL**	`/pic/mrp`, `/requisitions`, `/sourcing`, `/contracts`, `/purchase-orders`, `/receiving`, `/invoices`, `/spend`, `/admin` (45 pages). 57 of 58 screens are present. **SPD-08 Savings & Pipeline Tracker has no screen.** CLM-02 and INV-02 sit inside detail pages without their ID.
FE-2	AlphaHelion tokens; shared status chips and **lifecycle strip**	**PARTIAL**	Tokens and `StatusChip` are present. **No lifecycle strip.** The header shows a hard-coded "Manufacturing template" for every tenant (`fe/components/pic/WorkspaceShell.tsx:183`), which is a bug for CON/ITS/LOG/HLT/GEN.
FE-3	Object page: header, tabs (Lines · Approvals · Documents · Lineage · Audit), action bar; commands carry idempotency keys	**PARTIAL**	`fe/components/pic/object.tsx:ObjectPage` has Lines, Process, Lineage and Audit tabs plus a TaskPanel, used by 12 pages. **No generic Approvals or Documents tabs.** The client sends an Idempotency-Key on every POST.

ID	Item	Status	Evidence / gap
FE-4	Supplier portal as a separate Next.js app with supplier scopes	**PARTIAL**	Portal is `/pic/portal` **inside the same app**, restricted by the SUPPLIER_USER role (deviation; project DD-09 planned a second app)
FE-5	BPMN viewer (bpmn-js) in the process tab showing the live token	**NOT DONE**	`ProcessPath` renders a step list; `bpmn-js` is not a dependency

TD §17 Testing strategy

ID	Layer	Status	Evidence / gap
TST-U	Unit: 85% coverage on domain logic (Jest/Pytest)	**PARTIAL**	357 pytest functions (389 cases) in `tests/`. **No coverage measurement or gate** in `ci.yml`. **No frontend unit tests.**
TST-C	Contract: Pact, schema registry compatibility, must pass to merge	**NOT DONE**	None
TST-I	Integration: workflow + services + DB with RLS (nightly)	**DONE**	pytest against PG 16 with RLS on every push (`ci.yml`). Camunda test N/A.
TST-E2E	E2E-01..n through the UI with Playwright (release gate)	**NOT DONE**	No Playwright suite in the repo. The E2E-01 chain is exercised only as backend tests (`tests/test_signature_case.py`). The test workbook /03 has not been supplied.
TST-SEC	Security: RLS, SoD, authZ, OWASP/ZAP (release gate)	**PARTIAL**	`test_controls.py` and `test_review_platform.py`. ZAP is report-only.
TST-PERF	Performance: k6, cube benchmarks per release	**PARTIAL**	k6 scripts and the `perf_mrp` / `volume_gen` harnesses exist. Only the MRP run is recorded, and it failed.
TST-AI	AI evaluation before autonomy increases	**DONE**	`tools/ai_eval.py`, D9 sets, `tests/test_datasets_ai_eval.py`. Two gates fail (see ML-1, ML-3).

TD §18 Delivery plan & Definition of Done

ID	Item	Status	Evidence / gap
PH-0	Foundation exit: approval flow E2E; RLS suite green	**DONE**	`tests/test_controls.py`, `test_workspace_api.py`
PH-1	MVP P2P exit: E2E-01 manual PR path + idempotency tests, **with SAP/Tally connector**	**PARTIAL**	Backend E2E and idempotency tests pass. The SAP connector is a simulator; Tally is tested on mocks only.
PH-2	Source & Contract exit: RFQ-2231 → SA-5520 → PO inheritance	**DONE**	`tests/test_signature_case.py` ; DB PO-77410 with contract SA-5520
PH-3	Plan & Predict exit: MR-1043 netting; agent autonomy tests	**DONE**	MR-1043 = 9,000; agent guard and kill-switch tests (`test_intelligence_modules.py`)

ID	Item	Status	Evidence / gap
PH-4	Industry packs exit: industry test sheets green	**PARTIAL**	Five pack test files pass (`test_datasets_industry_*.py`). The customer's industry test sheets (workbook /03) have not been supplied.
DOD-1	Acceptance criteria and linked test cases pass; screen matches blueprint ID and states	**PARTIAL**	No mapping to the test workbook (not supplied); 57/58 screens
DOD-2	Audit and domain events emitted **and documented in the schema registry**	**PARTIAL**	Events are emitted; no registry, no event schema docs
DOD-3	Rules externalised (no hard-coded thresholds); SoD checks covered by tests	**PARTIAL**	Residual hard-codes (TD-1.1f); SoD tests DONE
DOD-4	OpenAPI updated; idempotency and concurrency handled; problem+json	**PARTIAL**	Idempotency on 26/138 POSTs; ETag only on POs
DOD-5	WCAG 2.1 AA check and performance budget met	**NOT DONE**	No accessibility audit; NFRs not measured or failed
DOD-6	Runbook and dashboards updated for each new failure mode	**PARTIAL**	Runbooks DONE; **no dashboards** (only a ServiceMonitor)

TD §19 Appendix

ID	Item	Status	Evidence / gap
APX-1	PR line payload (itemType FREE_TEXT, materialGroup, priceEstimate, accountAssignment, purchaseType, reusable, mdmRequestId)	**DONE**	`pr_line` columns match; `api/plan_request.py` item_type pattern
APX-2	PO DDL (status CHECK, rule_set_version, risk_score, touchless, total_minor, UNIQUE tenant+number+version)	**DONE**	`ck_purchase_order_status` ; `purchase_order` columns
APX-3	Audit event (onBehalfOf, before/after, ruleSetVersion, aiRef, prevHash, hash)	**DONE**	`audit_event` columns

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Part 2 — Product Blueprint

BP §2.1 Fourteen-stage lifecycle

ID	Stage	Status	Evidence / gap
LC-1/2	Demand planning; MRP run	**DONE**	MRP-01..04 (`fe/app/pic/mrp/**`), `svc/forecasting.py` , `svc/planning.py`
LC-3/4	Purchase requisition; guided buying	**DONE**	REQ-01..04, `svc/requisition.py:search` (P03 order)
LC-5/6	Strategic sourcing; supplier selection under capacity and risk	**PARTIAL**	Capacity is DONE; risk constraints are missing (ALG-9.3)
LC-7	Contract management	**DONE**	CLM screens; `svc/contract.py` , `svc/amendments.py`
LC-8/9	Purchase order; supplier fulfilment	**DONE**	PO-01..04, RCV-01
LC-10	GR / service entry	**DONE**	RCV-02..04
LC-11/12	Invoice matching; finance integration	**PARTIAL**	Matching is DONE; real ERP posting is simulator-only (INT-SAP)
LC-13	Spend analytics	**PARTIAL**	Built on Postgres; scale not proven (AD-04)
LC-14	Predictive intelligence	**PARTIAL**	Heuristic models (TD §13)

BP §3 Core principles P01–P14

ID	Principle	Status	Evidence / gap
P01	Traceable purpose: purpose code, requester, cost object, origin on PR/PO/contract; no submit without purpose	**DONE**	`requisition.purpose_code` , `purchase_order.purpose_code` ; submit guard in `svc/requisition.py`
P02	Two demand origins, stored and never overwritten	**DONE**	Trigger `pic.protect_requisition_origin` (migration 0002)
P03	Reuse before create: contract → catalogue → material → free text	**DONE**	`svc/requisition.py:search` (l.82–114) channel order; lines need the search result (l.183, 191). The Guided Buying **agent** is not wired (see AG-03).
P04	Free text never auto-creates a material	**DONE**	DB CHECK on pr_line; promotion only via `promote-material`
P05	Governed material creation: request → duplicate check → classification → stewards → ERP sync	**DONE**	F9; `svc/mdm.py`
P06	Evidence-based sourcing: approved requirements, qualified invitees, published criteria, authorised award	**DONE**	`svc/sourcing.py` (G-2, G-3, award approval)

ID	Principle	Status	Evidence / gap
P07	PO inherits PR, sourcing, contract, price, Incoterms, terms, tolerances; overrides flagged and justified	**DONE**	`svc/order.py:_create_po` (`inherited` , `overrides` , P07 rule violations)
P08	Purchase-type rules (7 types)	**DONE**	`PURCHASE-TYPE-MATRIX` rule; D2 cases (see BP §22)
P09	Approvals and SoD on awards, contracts, master data, payments	**PARTIAL**	Approval chains and SoD are DONE. The SoD mitigation path is missing (SEC-SOD).
P10	Complete lineage, including integration history	**DONE**	`document_link` , `integration_message` , Lineage and Audit tabs
P11	Explainable, governed AI: evidence, confidence, model version, permission; AI never commits spend	**DONE**	Answer contract (CP-1); agent runs log model version and evidence
P12	Three kinds of savings; only realised reaches P&L	**DONE**	`savings_ledger_entry` types NEGOTIATED / AVOIDANCE / REALISED with a `counts` flag
P13	Exactly-once ERP posting: idempotency keys, outbox, reconciliation	**PARTIAL**	Keys and outbox are DONE. Reconciliation is manual, not scheduled (INT-12.2-5). Only the simulator has been proven.
P14	Configurable by tenant, company code, plant, category, purchase type	**PARTIAL**	Rule scope is DONE. Configurable mandatory fields and editable workflows are missing (G-1, F-ver).

BP §4 Roles and §4.1 SoD

ID	Item	Status	Evidence / gap
ROLE-1..15	15 roles: Requester, Cost-object owner/Approver, MRP controller, Buyer, Category manager, Supplier manager, Contract manager/Legal, Master-data steward, Receiver, Service owner, AP clerk, Finance controller, CPO/Procurement head, Supplier user, Tenant admin	**DONE**	`plat/auth.py:ROLE_SCOPES` has 20 roles covering all 15, plus DEPT_HEAD, LEGAL, QUALITY_INSPECTOR and CAPEX_COMMITTEE. Seeded users in every tenant.
ROLE-ASSIGN	SoD also checked at role assignment	**DONE**	`svc/admin.py:21` incompatible role pairs
SOD-01	Create supplier master × create PO for the same supplier (High)	**DONE**	`seed/reference.py:SOD_RULES` ; `sod.guard("PO_CREATE")`
SOD-02	Change bank details × approve payment / post invoice (Critical)	**DONE**	SOD-02 and SOD-02a; PAYMENT_APPROVE and INVOICE_POST guards
SOD-03	Create PO × post GR for the same PO (High)	**DONE**	GR_POST guard
SOD-04	Create PO × approve invoice exception for the same PO (High)	**DONE**	INVOICE_EXCEPTION_APPROVE guard
SOD-05	Award event × score the same event as sole evaluator (Medium)	**DONE**	`plat/sod.py:_sole_evaluator`
SOD-06	Request material × approve as steward (Medium)	**DONE**	MATERIAL_STEWARD_APPROVE guard
SOD-07	Initiate savings × validate the same savings (High)	**DONE**	SAVINGS_VALIDATE guard
SOD-AGT	An agent inherits the SoD rules of its role	**PARTIAL**	Holds only through the underlying commands; the agent guard has no SoD check (AGT-2)

BP §6.1 Screen design conventions

ID	Convention	Status	Evidence / gap
UX-1	Left nav grouped by lifecycle; header with tenant, industry template and Copilot; **lifecycle strip in the footer**	**PARTIAL**	Nav groups (Decide, Plan & request, …) in `WorkspaceShell.tsx`. **The template label is hard-coded "Manufacturing". No lifecycle strip.**
UX-2	Object pages: header, tabs (Lines · Approvals · Documents · Lineage · Audit), action bar	**PARTIAL**	See FE-3
UX-3	Worklists: filter chips, saved views, SLA badges, bulk actions, side panel	**PARTIAL**	Lists and task SLA due dates exist. **No saved views, and no bulk actions** were found in `fe/app/pic`.
UX-4	AI surfaces: inline recommendation with "Explain"; accept/reject logged	**DONE**	Recommendations with feedback (`POST /v1/recommendations/{id}/feedback`), agent console
UX-5	State colours green/amber/red/blue	**DONE**	`StatusChip`
UX-6	WCAG 2.1 AA, keyboard-complete, colour never the only signal	**NOT DONE**	No audit (NFR-A11Y)

BP §7 M01 Spend & Category Intelligence

ID	Item	Status	Evidence / gap
M01-C1	Multi-ERP spend cube (PO, invoice, contract, consumption)	**PARTIAL**	Cube built from posted invoices plus ERP history only (`svc/savings.py:_spend` , `spend_cube`). **No PO commitments, consumption or multi-ERP source tagging.**
M01-C2	Configurable taxonomy (UNSPSC or own) with ML auto-classification and steward review	**PARTIAL**	Taxonomy nodes (25–30 per tenant) and UNSPSC mapping (56); steward queue below 0.80. Classification is rule-based, not ML (ML-1).
M01-C3	Addressable / contracted / non-contracted / unmanaged	**DONE**	`svc/spend.py:overview` (addressable, managed, contracted, maverick)
M01-C4	Duplicate-supplier and fragmented-purchase detection	**DONE**	SPD-03 clusters (`svc/spend.py:433+`); SPLIT_PO maverick rule
M01-C5	Concentration, price variance, off-contract, aggregation opportunities	**PARTIAL**	HHI, off-contract and opportunities exist. **The cube has no price-variance or aggregation measure.**
M01-C6	Category strategy, annual plan, category budgets	**DONE**	`category_plan` (15), `POST /v1/category-plans` , CATEGORY-BUDGET check at requisition

ID	Item	Status	Evidence / gap
M01-C7	Sourcing pipeline with forecast vs realised savings and cost avoidance	**PARTIAL**	Savings register/waterfall exist; **no SPD-08 pipeline tracker**
M01-C8	Actual vs budget, contract, historical price, benchmark	**PARTIAL**	Budget and contract compliance exist; **no external benchmark**
M01-C9	Spend forecast by category, supplier, plant, period	**DONE**	`GET /v1/spend/forecast` (80% band, MAPE 0.035)
M01-D	Cube dimensions (cat L1–L4, supplier parent/child, plant, cc, cost centre, WBS, material group, period, currency doc/group)	**PARTIAL**	Dimensions: category, supplier, plant, company_code, period, material_group, cost_object. **No supplier parent/child, and no document/group currency.**
M01-R1	Spend de-duplicated across PO and invoice via lineage	**DONE**	Spend rows come only from posted invoices, so there is no double count
M01-R2	Classification < 0.80 confidence → steward queue	**DONE**	SPEND-GOVERNANCE `classificationThreshold 0.80`; `classification_review` (67)
M01-R3	Intercompany excluded from addressable spend	**DONE**	`svc/spend.py:_apply_addressable` (IC-TRANSFER)
M01-R4	Maverick rules tenant-configurable; each hit has an owner and disposition	**DONE**	`maverick_rule` (9 codes per tenant, incl. PO_AFTER_INVOICE, SPLIT_PO, NON_CATALOGUE, OFF_CONTRACT_PRICE, ONE_TIME_VENDOR); `POST /v1/spend/maverick-hits/{id}/disposition`; 446 hits
M01-S	Screens SPD-01..09	**PARTIAL**	`fe/app/pic/spend/page.tsx` tabs SPD-01..07 and 09; **SPD-08 missing**
M01-K	KPIs: spend under management, contract coverage, maverick %, classification accuracy, realisation rate	**PARTIAL**	The first four are computed (`svc/spend.py`, `svc/insight.py`). **Savings realisation rate % is not computed.**

BP §8 M02 Demand Planning & MRP

ID	Item	Status	Evidence / gap
M02-C1	Demand inputs: forecast, customer orders, project/WBS, reorder points, consumption	**PARTIAL**	`demand_element.type` FORECAST / SALES_ORDER / RESERVATION / WBS_DEMAND; forecasts from consumption. **No reorder-point (Kanban) planning.**
M02-C2/3	Netting formula; lot sizing incl. pack rounding	**DONE**	ALG-9.1
M02-C4	Lead-time scheduling with PDT, GR time and calendar	**DONE**	`_Reads.release_date`
M02-C5	Exception messages (reschedule in/out, cancel, past due, below safety)	**DONE**	`svc/planning.py:net_material`

ID	Item	Status	Evidence / gap
M02-C6	Planner workbench with firming and conversion	**DONE**	`POST /v1/planned-orders/{ref}/firm`, `/convert`; MRP-02
M02-C7	Simulation before live run; in-platform or ERP MRP	**DONE**	Mode SIMULATION (MR-1044 SIMULATED); ERP import (MR-1048)
M02-C8	Source list and quota applied at conversion	**DONE**	`svc/source.py:determine`
M02-D	Run modes net-change (default), regenerative, simulation	**DONE**	`planning.run` validates the mode
M02-R1	Only materials with MRP type active are planned	**DONE**	`Material.mrp_type == "PD"` (`svc/planning.py:282`)
M02-R2	Firmed planned orders not changed by later runs	**DONE**	RESCHEDULE messages instead of changes
M02-R3	Conversion inherits material, qty, date, plant, sloc, source	**DONE**	`svc/planning.py:convert` (l.399+)
M02-R4	Simulation never creates documents	**DONE**	`mode != "SIMULATION"` guards (l.298, 322, 342)
M02-S	Screens MRP-01..04	**DONE**	`fe/app/pic/mrp/page.tsx`, `mrp/[ref]`, `mrp/forecast`
M02-K	KPIs: MAPE, exception backlog, PO conversion lead time, stock-outs, days of cover	**PARTIAL**	Only MAPE (`svc/forecasting.py`). **The other four are not computed.**
M02-NFR	50,000 materials < 10 min	**NOT DONE**	NFR-MRP (681 s best)

BP §9 M03 Master Data Governance

ID	Item	Status	Evidence / gap
M03-C1	New material request with mandatory classification	**DONE**	`svc/mdm.py`, MDM-01
M03-C2	Duplicate detection: fuzzy description, MPN, attributes	**DONE**	MDM-DUP weights; F9-04
M03-C3	Steward work queues by domain (purchasing, quality, finance, plant)	**DONE**	4 parallel domain tasks; MDM-02
M03-C4	Field-level ownership and approval by material type	**PARTIAL**	Domain-level approval only. **No per-field ownership by material type.**
M03-C5	**Material** change requests with before/after and impact on open POs, contracts, catalogues	**NOT DONE**	Only supplier master changes exist (`svc/master_changes.py`). No change request for an existing material.
M03-C6	Supplier master: registration, bank verification, tax IDs, sanctions, duplicate check	**DONE**	F8; MDM-04 tax/address/contact (`svc/master_changes.py`, two approvers, never the requester)

ID	Item	Status	Evidence / gap
M03-C7	ERP sync with status and error handling	**PARTIAL**	Status, retry and dead-letter are DONE. The target is the SAP simulator only (INT-SAP).
M03-C8	Golden-record history and audit	**DONE**	Audit trail; F9-08 "golden record live"
M03-R1	Free-text line promotable; PR keeps its line; material linked after approval	**DONE**	`promote-material` route; `pr_line.linked_material_id`
M03-R2	Bank changes: two finance approvers, never the requester	**DONE**	SEC-BANK
M03-R3	Blocked at ≥ 0.92 unless a steward overrides with reason	**DONE**	`POST /v1/materials/requests/{ref}/override`
M03-R4	Sync to ERP only after final approval	**DONE**	`svc/mdm.py:251` (SYNCED_TO_ERP after APPROVED)
M03-S	Screens MDM-01..04	**DONE**	`fe/app/pic/materials/**`, `suppliers/[code]/master-data.tsx`
M03-K	KPIs: request cycle time, duplicate rate, first-time-right, bank-fraud attempts blocked	**NOT DONE**	None computed (bank-change alerts exist as risk signals)

BP §10 M04 Requisitioning & Guided Buying

ID	Item	Status	Evidence / gap
M04-C1	Intake "What do you need?" across contracts, catalogues, materials, policy	**DONE**	`GET /v1/guided-buying/search`
M04-C2	Hosted catalogues and punch-out (cXML/OCI)	**DONE**	`catalogue_item`; punch-out (INT-PUNCH; real suppliers EXTERNAL)
M04-C3	Free-text and service requests **with templates** (services, IT, marketing, facilities)	**PARTIAL**	Free text and SERVICE lines exist; **no request templates**
M04-C4	Account assignment: cost centre, WBS, internal order, asset, **sales order**	**PARTIAL**	COST_CENTER / WBS / ORDER / ASSET (`models/platform.py:63`); **no sales-order account assignment**
M04-C5	Budget check (hard/soft) and commitment	**DONE**	BUDGET rule; `budget_commitment` (33)
M04-C6	Policy guidance: preferred supplier, required quotes by band, channel	**DONE**	QUOTE-POLICY (each quote needs supplier, reference and date); `preferred_supplier_id`

ID	Item	Status	Evidence / gap
M04-C7	Approval chain by value, category, cost object, risk	**DONE**	DMN-01, one task per cost object
M04-C8	Converted MRP PRs in the same buyer queue	**DONE**	`GET /v1/buyer-workbench` (REQ-04)
M04-R1	Search order enforced	**DONE**	P03
M04-R2	Free text above the configurable value needs 3 quotes or a sourcing event	**DONE**	QUOTE-POLICY `oneQuoteBelow 2 L`
M04-R3	Budget hard stop on capex; soft stop with justification on opex	**DONE**	BUDGET `hardFor ASSET,WBS` , `softFor COST_CENTER,ORDER`
M04-R4	PR changes after approval restart approval only on value/account/supplier change	**DONE**	`svc/requisition.py` reapprove (code review fix 7)
M04-S	Screens REQ-01..04 (approval inbox mobile-friendly)	**DONE**	`fe/app/pic/requisitions/**` , `tasks` , `buyer` . Mobile-friendly only through responsive layout; not tested.
M04-K	KPIs: catalogue/contract channel %, PR approval cycle time, free-text %, budget-blocked PRs	**PARTIAL**	Only PR approval cycle (`svc/insight.py` avgApprovalHours)

BP §11 M05 Strategic Sourcing

ID	Item	Status	Evidence / gap
M05-C1	Event types RFI, RFQ, RFP, reverse auction, multi-round, **lot/item bidding**	**PARTIAL**	RFI/RFQ/RFP/REVERSE_AUCTION types (`api/source_contract.py:44`); rounds and live auction with soft close (`svc/auction.py`). **One item per event, so no lot/item bidding.** RFI/RFP share the RFQ flow (no technical envelope).
M05-C2	Requirement bundling from PRs, MRP demand or category plan	**PARTIAL**	One requisition line per event; **no bundling**
M05-C3	Invitation from qualified list with segment and risk filters	**PARTIAL**	Eligibility is enforced; `svc/sourcing_strategy.py` recommends invitees. **No segment or risk filter UI.**
M05-C4	Weighted scoring: price/TCO, quality, delivery, technical, ESG, risk	**PARTIAL**	Criteria are configurable per event; **no ESG or risk score source**
M05-C5	Should-cost and benchmarks; TCO components	**PARTIAL**	TCO is DONE; should-cost covers one commodity (ML-4)

ID	Item	Status	Evidence / gap
M05-C6	Award scenarios: lowest cost, best value, split, capacity-constrained	**DONE**	`svc/sourcing.py:build_scenarios`
M05-C7	Award approval and notification; outcome flows to contract or PO	**DONE**	AWARD-APPROVAL; `contract-from-award` consumer; notifications
M05-C8	Sealed bids, clarifications, Q&A log	**DONE**	Sealing (G-3); `sourcing_clarification` with endpoints
M05-R1	Invite only Qualified (or Conditional with waiver)	**DONE**	`_require_eligible`
M05-R2	Bids sealed until close; early access blocked and logged	**DONE**	G-3
M05-R3	Weights locked at publish	**DONE**	`weights_locked`
M05-R4	Non-lowest award needs written rationale	**DONE**	`svc/sourcing.py:558`
M05-R5	Award above threshold needs category head + finance	**DONE**	AWARD-APPROVAL ≥ ₹50 L: CATEGORY_MANAGER, CPO, FINANCE_CONTROLLER
M05-S	Screens SRC-01..05	**DONE**	`fe/app/pic/sourcing/**`
M05-K	KPIs: sourcing cycle time, competitive events %, negotiated savings %, bidders per event	**PARTIAL**	Negotiated savings is computed. The others appear only as list fields (bidders); **no cycle-time KPI**.

BP §12 M06 Supplier Lifecycle & Performance

ID	Item	Status	Evidence / gap
M06-C1	Self-registration and invitation; duplicate and sanctions screening	**DONE**	`svc/invitations.py` ; `/public/invitations/{token}/register` ; screening disposition gate
M06-C2	Qualification by category: questionnaires, certificates, audits, sample approval (PPAP)	**DONE**	SUPPLIER-QUAL (`minQuestionnaireScore` , `minAuditScore` , `sampleApprovalRequired`)
M06-C3	Status model per category	**DONE**	SM-SUP
M06-C4	Segmentation (strategic, leverage, bottleneck, routine) and risk tiering	**PARTIAL**	Kraljic lives on the category plan; `supplier.segment` and `risk_tier` are plain fields. **No supplier segmentation logic.**

ID	Item	Status	Evidence / gap
M06-C5	Scorecards: OTIF, PPM, responsiveness, invoice accuracy, ESG	**PARTIAL**	OTIF, PPM and invoice accuracy (`scorecard` , 40 rows). **No responsiveness, no ESG.**
M06-C6	Risk monitoring: financial, geographic, compliance, single-source, cyber	**PARTIAL**	Financial tier, compliance (sanctions, certificates), delivery and bank-change signals. **No geographic, single-source or cyber signals.**
M06-C7	Corrective action and development plans	**PARTIAL**	CAPA open → respond → verify (`/capas`). **No development plans.**
M06-C8	Supplier portal: profile, documents, RFQs, POs, ASNs, invoices	**DONE**	`fe/app/pic/portal/page.tsx` , `/portal/profile`
M06-D	SLM-01 reads twin capability, evidence, free capacity; scorecards write back	**PARTIAL**	INT-TWIN (local table, no Twin connector)
M06-R1	Only Qualified/Preferred receive POs unless one-time-vendor policy	**PARTIAL**	Not-qualified suppliers add risk points and block touchless; one-time vendor is flagged as maverick. **There is no hard block on PO release** to a not-qualified supplier (only Blocked is refused).
M06-R2	Expired mandatory certificate → Conditional automatically	**DONE**	G-9
M06-R3	Blocking stops new POs; open POs flagged for buyer decision	**DONE**	`svc/supplier.py:block` (`openPosFlagged` , payment block)
M06-R4	Scorecards recompute **monthly** and on every GR/invoice event	**PARTIAL**	Event-driven recompute is DONE (consumers). **No monthly job.**
M06-S	Screens SLM-01..05	**DONE**	`fe/app/pic/suppliers/**` , `portal`
M06-K	KPIs: onboarding cycle time, qualified coverage %, OTIF, PPM, high-risk %	**PARTIAL**	Qualified coverage, OTIF, PPM and high-risk are computed; **onboarding cycle time is not**

BP §13 M07 Contract Lifecycle Management

ID	Item	Status	Evidence / gap
M07-C1	Contract types: master agreement, quantity, value, scheduling agreement, rate card, SOW, NDA	**PARTIAL**	The create API accepts only SCHEDULING_AGREEMENT, QUANTITY_CONTRACT and VALUE_CONTRACT (`api/governance.py:34`). Rate card and SOW are modelled as value-contract lines (ITS). **No master agreement or NDA type.**
M07-C2	Clause library with fallback positions and risk scores	**DONE**	CLM-PLAYBOOK; `clause` (104 rows)

ID	Item	Status	Evidence / gap
M07-C3	Templates by category and **jurisdiction**; AI redline vs playbook	**PARTIAL**	Templates by type; deterministic redline. **No jurisdiction dimension, and no LLM** (ML-8).
M07-C4	Approval workflow and e-signature	**EXTERNAL**	Approval is DONE. The e-sign provider adapter (`svc/contract.py:576` , `/esign/webhook`) needs `PIC_ESIGN_*` ; until then signatures are internal with a document hash.
M07-C5	Obligations, milestones, price indexation, **rebates**	**PARTIAL**	Obligations (34) and index resets are DONE. **No rebates.**
M07-C6	Consumption tracking; alerts at 80/90/100%	**DONE**	`GET /v1/contracts/{ref}/consumption` ; thresholds
M07-C7	Renewal, expiry, termination	**DONE**	Timers; amendments (`svc/amendments.py` : variation, extension, termination)
M07-C8	Contract compliance: off-contract buying and price leakage	**DONE**	`GET /v1/spend/compliance` ; OFF_CONTRACT and OFF_CONTRACT_PRICE rules
M07-D	Indexation creates a new price-validity period each reset	**DONE**	`POST /v1/contracts/{ref}/index-resets` ; `price_validity` (35)
M07-R1	Only executed contracts referenced by POs	**DONE**	Source determination uses ACTIVE contracts with published validities
M07-R2	PO price = contract price ± tolerance, or flagged	**DONE**	TD-5.2b
M07-R3	Contract above threshold needs legal + finance	**DONE**	CONTRACT-APPROVAL ≥ ₹1 Cr: LEGAL, FINANCE_CONTROLLER, CPO
M07-R4	Indexation never overwrites	**DONE**	New `price_validity` rows
M07-S	Screens CLM-01..04	**DONE**	`fe/app/pic/contracts/**` (CLM-02 inside `[ref]`)
M07-K	KPIs: contract cycle time, leakage, expiring in 90 days, obligation on-time %	**PARTIAL**	Leakage and renewals calendar (`/contracts-renewals`); **no cycle-time or obligation on-time %**

BP §14 M08 Operational Purchasing

ID	Item	Status	Evidence / gap
M08-C1	Automatic PO from PR with source determination	**DONE**	`svc/order.py:create_from_requisition` + `svc/source.py`
M08-C2	Manual PO, release order against contract, scheduling-agreement schedule lines	**DONE**	`po_schedule_line` (78), `PUT /purchase-orders/{ref}/lines/{n}/schedule`
M08-C3	AI PO validation: price vs contract/history, supplier risk, split detection, risk score	**PARTIAL**	Contract-price deviation, split and risk are DONE. **No price check against history (last price).**
M08-C4	Approval strategy by value, category, risk, deviation	**DONE**	PO-APPROVAL bands + deviation chain + change approval
M08-C5	Transmission (portal, email, EDI/cXML) and confirmation	**EXTERNAL**	Portal is DONE. EDI/e-mail adapters need the gateway and SMTP. **cXML OrderRequest is not built.**
M08-C6	Change orders with versioning and re-confirmation	**DONE**	G-6
M08-C7	Expediting with late-delivery probability	**PARTIAL**	`/expediting`, EXPEDITING agent. The probability comes from the seeded twin, not a model (ML-5).
M08-C8	Touchless PO for low-risk lines	**DONE**	DMN-02; AXN touchless 9/23 = 39%
M08-R1	PO price defaults from contract; deviation > tolerance needs justification	**DONE**	P07
M08-R2	Not released to a Blocked supplier	**DONE**	G-5
M08-R3	Split to dodge a threshold is detected and escalated	**DONE**	`svc/order.py:_split_pattern` (same supplier, material group and cost object within 7 days, below the ₹2 L band) → 25 risk points, not touchless, routed to approval
M08-R4	Qty/price/date change after confirmation → new version and re-confirmation	**DONE**	G-6
M08-S	Screens PO-01..04	**DONE**	`fe/app/pic/purchase-orders/**`
M08-K	KPIs: touchless %, PO cycle time, confirmation within SLA, change rate	**PARTIAL**	Touchless, confirmation SLA and change rate exist; **no PO cycle time**

BP §15 M09 Fulfilment, GR & Service Entry

ID	Item	Status	Evidence / gap
M09-C1	ASN from portal/EDI with packing and batch data	**DONE**	`POST /v1/asns`, EDI DESADV
M09-C2	GR for stock, non-stock, asset; partial receipts	**DONE**	`svc/receiving.py`
M09-C3	Quality inspection (sampling, usage decision), returns, replacements	**DONE**	ISO 2859-1 sample table; `/inspection-lots/{ref}/usage-decision`; `/return-deliveries/{ref}/replacement`
M09-C4	SES: milestones, T&M, unit-rate, progress %	**DONE**	`svc/receiving.py:create_ses(method=…)`; CON 35% progress; ITS 320 h
M09-C5	Receipt tolerances per material/supplier	**PARTIAL**	Tolerance is per contract or the default RCV-TOLERANCE rule. **No per-material or per-supplier tolerance.**
M09-C6	Consignment and subcontracting receipts	**DONE**	Movements 201K/411K, 541/543 (D2)
M09-C7	Batch, serial, expiry capture	**DONE**	Lot trace and recall (`/lots/{material}/{batch}/trace`, `/recall`; HLT)
M09-R1	SES approved by someone other than supplier and PO creator	**DONE**	G-7
M09-R2	Over-delivery above tolerance blocked	**DONE**	TD-5.2c
M09-R3	Quality-managed GR posts to quality stock until usage decision	**DONE**	F6-05 "yes: quality stock"
M09-R4	Expense receipt-less only below configured value	**DONE**	`receiptlessBelowMinor` (₹25k default, ₹1 L GEN)
M09-S	Screens RCV-01..04	**DONE**	`fe/app/pic/receiving/**`
M09-K	KPIs: receipt-to-PO accuracy, dock-to-stock, SES approval time, quality rejection rate	**NOT DONE**	None computed

BP §16 M10 Invoice Matching & Finance Integration

ID	Item	Status	Evidence / gap
M10-C1	Capture: portal, PDF/OCR, e-invoice (IRN), EDI	**PARTIAL**	Portal is DONE. OCR adapter (`svc/ocr.py` , needs `PIC_OCR_URL`) and IRN (INT-GST) are EXTERNAL. **No EDI INVOIC.**
M10-C2	2-, 3-, 4-way match; ERS	**DONE**	ALG-9.4 (ERS for consignment only; **not for scheduling agreements**, which the manufacturing template asks for)
M10-C3	Tolerances by amount and %, price and qty, per company code	**PARTIAL**	DMN-03 resolves per company code. **"Per supplier group" is not a scope dimension.**
M10-C4	Exception workflow: price, quantity, missing GR, tax, duplicate	**DONE**	`invoice_exception` (22); `/exceptions/{id}/resolve`
M10-C5	Credit/debit notes and retention	**DONE**	`/credit-notes` ; retention withhold and release (CON)
M10-C6	Posting with idempotency and reconciliation	**PARTIAL**	Idempotency is DONE; reconciliation is manual; simulator only
M10-C7	Payment status, early-payment discount, **supply-chain finance hooks**	**PARTIAL**	Payment runs, ISO 20022 files (`svc/bank_files.py`) and discount capture are DONE. **No SCF hooks.** Bank host-to-host is EXTERNAL.
M10-C8	GRIR accruals	**DONE**	`GET /v1/grir`
M10-R1	Duplicate check supplier + no. + amount + date (± tolerance)	**DONE**	TD-5.2d
M10-R2	Goods invoices need GR before approval (3-way) unless ERS	**DONE**	`svc/payables.py:match`
M10-R3	Service invoices match approved SES	**DONE**	match type SES
M10-R4	Idempotency key = tenant + company code + supplier + invoice no.	**DONE**	`svc/payables.py:posting_key`
M10-R5	Payment block until exceptions resolved	**DONE**	`payment_block` ; F7-04 "outside tolerance → payment block"
M10-S	Screens INV-01..04	**DONE**	`fe/app/pic/invoices/**` (INV-02 in `[id]`)
M10-K	KPIs: first-pass match, exception cycle time, duplicates prevented, discount capture, DPO	**PARTIAL**	First-pass match, discount capture and DPO (`/payables/insights`); **no exception cycle time or duplicates-prevented count**

BP §17 M11 Savings & Value Realisation

ID	Item	Status	Evidence / gap
M11-C1	Savings types negotiated / avoidance / realised	**DONE**	Ledger types
M11-C2	Baselines: last price, weighted average, budget, should-cost	**DONE**	`svc/savings.py:BASELINE_METHODS` (LAST_PRICE, WAVG_12M, STANDARD, SHOULD_COST)
M11-C3	Lifecycle Identified → … → Validated	**DONE**	SM-SAV
M11-C4	Finance validation and P&L/budget view	**PARTIAL**	Validation with SOD-07 is DONE. **No P&L/budget impact view.**
M11-C5	Leakage: negotiated not realised (off-contract, volume shortfall, price drift)	**DONE**	Waterfall leakage bars (`/savings/{ref}/waterfall`)
M11-D	Signature: negotiated 0.99 Cr; avoidance 0.32 Cr; realised YTD 0.38 Cr	**PARTIAL**	DB SV-0412: negotiated 0.9936 Cr and avoidance 0.318 Cr match. **Realised is 0.0129 Cr (₹1.29 L), not 0.38 Cr**: the seed posts too few invoices against the initiative.
M11-R1/2/3	Realised only from posted invoices; avoidance never counts; validator ≠ owner	**DONE**	ALG-9.5
M11-S	Screens SAV-01/02	**DONE**	`fe/app/pic/savings/page.tsx` (register + waterfall)
M11-K	KPIs: negotiated, realised, realisation rate, leakage	**PARTIAL**	Realisation rate is not computed

BP §18 M12 Predictive Intelligence & Agents

ID	Item	Status	Evidence / gap
M12-C1	Copilot with answer contract	**DONE**	CP-1..4 (deterministic, no LLM)
M12-C2	Predictions: **price trend**, late delivery, supplier risk, demand, invoice exception	**PARTIAL**	Demand forecast and rule-based risk/exception exist. **No price-trend model; late-delivery is seeded data.**
M12-C3	Should-cost by commodity	**PARTIAL**	ML-4
M12-C4	Agents L0–L4 with kill switch	**DONE**	`PUT /v1/agents/{code}/autonomy` , `/kill-switch`
M12-C5	Model registry, versioning, drift monitoring, feedback loop	**PARTIAL**	Version strings and feedback exist. **No registry, no drift monitoring.**

ID	Item	Status	Evidence / gap
M12-R1	Agents act only with their role's permissions	**DONE**	Agent principal is scoped to `run_as_role`
M12-R2	L3/L4 only for configured low-risk categories and values	**PARTIAL**	Value limit (₹25 L) and category list exist. Defaults have **no category restriction** (`categories` empty) for the L3/L4 agents.
M12-R3	Every agent action logged with inputs, model version, outcome	**DONE**	AGT-3
M12-S	Screens AIX-01..03	**DONE**	`fe/app/pic/page.tsx` (command centre), `copilot`, `agents`
M12-K	KPIs: agent-assisted %, recommendation acceptance %, prediction accuracy, hours saved	**PARTIAL**	Acceptance rate per agent (`agents.acceptance_rate`) and eval accuracy (D9). **No agent-assisted % or hours saved.**

BP §19 M13 Platform, Workflow & Administration

ID	Item	Status	Evidence / gap
M13-C1	Org model	**PARTIAL**	TD-4a (no legal entity or purchasing org/group)
M13-C2	Rule engine for thresholds, mandatory fields, tolerances	**PARTIAL**	Mandatory fields missing (G-1); `POST /v1/admin/rules`
M13-C3	**BPMN workflow designer** and approval matrix	**PARTIAL**	Approval-matrix simulation is DONE (`svc/policy_sim.py` , `POST /v1/admin/rules/simulate`). **No workflow designer**; flows are code.
M13-C4	Roles, permissions, SoD rule set, conflict reports	**DONE**	`/admin/roles` , `/admin/sod` , `sod_violation`
M13-C5	Integration hub: connectors, mappings, monitoring, replay	**PARTIAL**	Monitoring and replay (`/integration/messages` , `/admin/connectors`) are DONE. **Field mappings are fixed in code, not configurable.**
M13-C6	Audit trail explorer and data retention	**PARTIAL**	Audit explorer is DONE; retention is report-only (SEC-PRIV)
M13-C7	Industry template packs	**DONE**	TD-4d
M13-R1	Rule changes versioned and effective-dated; in-flight documents keep their version	**DONE**	AD-07
M13-R2	Workflow models versioned; running instances complete on their version	**PARTIAL**	F-ver

ID	Item	Status	Evidence / gap
M13-S	Screens ADM-01..06	**DONE**	`fe/app/pic/admin/page.tsx` tabs, `processes`
M13-K	KPIs: integration success %, MTTR of integration errors, SoD conflicts open	**PARTIAL**	Posting success % and SoD violations exist; **no MTTR**

BP §20 Signature case (Axion, AXN) — values in pic_neon_build

ID	Stage	Status	Evidence / gap
SIG-1	MR-1043 net 9,000	**DONE**	`mrp_run` MR-1043 NET_CHANGE COMPLETED; netting note in the seed
SIG-2	REQ-2311 origin MRP from PLO-55821	**DONE**	`tests/test_signature_case.py`
SIG-3/4	RFQ-2231 bids 412/398/405, should-cost 391; award 60/40	**DONE**	`seed/axion.py:375–407`; award_decision (3)
SIG-5	SA-5520 scheduling agreement, LME quarterly, 60-day terms	**DONE**	CLM-PLAYBOOK awardDefaults (LME_AL ±3%)
SIG-6	PO-77410 risk 18/100 touchless	**DONE**	DB PO-77410 risk 18, touchless true
SIG-7	ASN-61208 confirmed in 6 h with batches and CoC	**DONE**	Seeded ASN
SIG-8	GR-90231: 5,400 received / 5,380 accepted / 20 rejected	**DONE**	DB gr_line sums 5400 / 5380 / 20
SIG-9	INV-TF-8812 → CN-TF-221 → posted, ERP doc 5100004421	**DONE**	DB INV-TF-8812 PAID, erp_doc_no 5100004421
SIG-10	SV-0412 negotiated 0.99 Cr / avoidance 0.32 Cr / realised 0.38 Cr	**PARTIAL**	Realised is only 0.0129 Cr (M11-D)
SIG-11	AIX-01 portfolio: spend 412 Cr, under management 78%, coverage 71%, maverick 6.2%, touchless 41%, first-pass 86%, realised 6.3 Cr	**PARTIAL**	DB AXN, 24 months: spend 803.9 Cr (≈ 402 Cr per year), coverage 69.7%, maverick 6.1%, touchless 39%, **first-pass 72% (13/18)**, realised ≈ ₹1.3 L. The demo storyline numbers are not met.

BP §21 BPMN flows (branch nuances beyond TD §8)

ID	Nuance	Status	Evidence / gap
BF-1	F1 simulation ends without documents; firmed protected; contract → quota → source list → info record	**DONE**	ALG-9.1/9.2
BF-2	F2 free text never creates materials; parallel MDM does not block the PR	**DONE**	P04; F2-05/06
BF-3	F3 weights lock; sealed; non-lowest rationale; savings recorded as negotiated	**DONE**	M05-R*; `savings-negotiated` consumer
BF-4	F4 indexation validity periods; alerts 80/90/100; renewal 120 d	**DONE**	M07
BF-5	F5 touchless criteria; changes outside tolerance create a new version back to the buyer	**DONE**	F5-08 outcomes in DB

ID	Nuance	Status	Evidence / gap
BF-6	F6 SES approver ≠ PO creator; over-delivery blocked; rejections feed scorecard and twin	**DONE**	G-7; `twin_feedback` (local, see INT-TWIN)
BF-7	F7 retries never double-post; realised savings only after posting	**DONE**	INT-12.2; ALG-9.5
BF-8	F8 per-category qualification; expired certificates downgrade	**DONE**	SM-SUP; G-9
BF-9	F9 high scores redirect; stewards in parallel	**DONE**	F9-04 "duplicate: requester redirected to DC-330"

BP §22 Purchase-type rule matrix (P08)

ID	Type	Status	Evidence / gap
PT-1	Stock: material required; GR mandatory, QI if quality-managed; 3-way	**DONE**	Matrix STOCK; STOCK needs a material (code review F2 #3)
PT-2	Non-stock: material optional; cost centre/order/WBS; GR (non-valuated); expensed at GR; 3-way	**DONE**	Matrix NON_STOCK
PT-3	Service: SES approval; 3-way vs SES	**DONE**	Matrix SERVICE (`ses: True`)
PT-4	Expense: GR optional below threshold; 2-way below, else 3-way	**DONE**	`receiptlessBelowMinor` ; D2 receipt-less case
PT-5	Capital asset: asset/WBS (capex); GR mandatory; capitalised; capex budget, board limits	**PARTIAL**	Asset or WBS required; capex committee; hard budget stop (D2). **No capitalisation in the ERP GR payload** (status doc open item), and no board-limit band.
PT-6	Consignment: GR to consignment stock; liability at withdrawal; ERS	**DONE**	201K/411K withdrawals; `settle_consignment` (D2, HLT)
PT-7	Subcontracting: material required plus components; GR of the finished part; component consumption; processing fee	**DONE**	`/purchase-orders/{ref}/components` ; 541/543/544 (D2). Job-work tax code SAC 9988 per line still needs a schema decision.

BP §23 Industry templates and §23.1 nuances

ID	Template	Status	Evidence / gap
IND-MFG	Manufacturing: MRP, should-cost RFQ, PPAP, scheduling agreements, quota, QI, 3-way, **ERS for scheduling agreements**, **Kanban/reorder for C-parts**, PPV	**PARTIAL**	AXN covers MRP, PPAP gate, quota and QI. **No ERS on scheduling agreements, no Kanban/reorder-point planning, no PPV against standard cost KPI.**

ID	Template	Status	Evidence / gap
IND-CON	Construction: WBS, BOQ, tender by trade, subcontract + rate contracts, measurement SES, retention, **advance recovery**, variation orders as amendments, **performance bank guarantee**	**PARTIAL**	CON tenant: SC-118, WBS C-2207, SES at 35/70/100%, 5% retention, VO-01 amendment. **Advance recovery is a rule flag nothing reads** (`PROJECT-CONTROLS.advanceRecovery` unused). The bank guarantee exists only as clause text and an approval comment.
IND-ITS	IT services: SOW, rate cards, timesheet/milestone SES, licence renewals, InfoSec assessment, **unused-licence flag**	**PARTIAL**	ITS: SOW-4410, RC-22, 320 h SES, renewal alert/extension; InfoSec in the qualification questionnaire. **No licence-utilisation tracking or unused-licence flag.**
IND-LOG	Logistics: lane tenders, rate contracts with fuel indexation, POD as service confirmation, freight audit, carrier compliance, **detention/demurrage**	**PARTIAL**	LOG: lane CHE→PNQ ₹54/km, 15 trips, POD, freight audit (FREIGHT-AUDIT `rateTolerancePct 0`), carrier certificates mandatory. `fuelIndex` **and** `accessorialsNeedPod` **are unused flags; no detention/demurrage rules.** The diesel index is demo data.
IND-HLT	Healthcare: par levels, VAC, GPO, consignment, lot/expiry/serial, **cold chain**, usage-triggered replenishment, recall, **clinical sign-off in award**, regulatory approvals	**PARTIAL**	HLT: KN-77 consignment, lot/expiry/serial, recall trace, usage-triggered PO, CDSCO/ISO 13485 mandatory certificates. `AWARD-APPROVAL.clinicalSignOff` **is never read, so clinical sign-off is only a comment. No cold chain, no par-level planning.**
IND-GEN	General: guided buying, quick quote/3-quote, catalogue, desktop receipt, receipt-less below threshold, 2-way/3-way, **P-card limits**, budget	**PARTIAL**	GEN: punch-out laptop ₹78,500 with one approval, receipt-less below ₹1 L. **No P-card.**
IND-KPI	Template KPIs (PPV, cost to complete, rate compliance, cost per tonne-km, expiry write-offs, catalogue adoption …)	**NOT DONE**	They are listed as labels in `seed/templates.py`; none is computed

BP §24 Approval matrix (Table 24) and §24.1 control points

ID	Band	Status	Evidence / gap
APR-1	Requisition < ₹25,000: auto (policy-compliant catalogue/contract)	**DONE**	DMN-01 band 1 `autoChannels`
APR-2	Requisition ₹25,000 – 2 L: cost-object owner	**DONE**	DMN-01
APR-3	Requisition 2 – 10 L: cost-object owner + department head	**DONE**	DMN-01
APR-4	Requisition > 10 L: + procurement head + finance controller; capex follows asset policy	**DONE**	DMN-01 + `assetAdds CAPEX_COMMITTEE`. Band-edge decision pending (see DMN-01).
APR-5	Sourcing award < 50 L: category manager; rationale if not lowest	**DONE**	AWARD-APPROVAL
APR-6	Sourcing award ≥ 50 L: category manager + CPO + finance; conflict-of-interest declaration	**DONE**	AWARD-APPROVAL; `svc/sourcing.py:553`

ID	Band	Status	Evidence / gap
APR-7	Contract ≥ 1 Cr or non-standard clauses: Legal + Finance + CPO; fallback deviations scored	**DONE**	CONTRACT-APPROVAL `above`; non-standard redlines route to Legal
APR-8	PO deviation: price > contract + 2% **or > last price + 5%** → buyer lead; justification mandatory	**PARTIAL**	Deviation is measured against the **contract's own price tolerance** (default 0.5%), not a +2% band. **The "last price + 5%" check for non-contract POs is not built.** Buyer lead = a second BUYER (PO-APPROVAL `deviationChain`).
APR-9	Supplier bank change: Finance ×2 with call-back; never the requester	**DONE**	SEC-BANK
APR-10	New reusable material: domain stewards; duplicate score < 0.92	**DONE**	F9
CTL-1	Budget check at PR submit (hard capex/WBS, soft opex)	**DONE**	BUDGET rule
CTL-2	Split-PO detection: same supplier + material group + cost object within 7 days above the band	**DONE**	`_split_pattern`
CTL-3	Price above contract tolerance needs justification and buyer-lead approval	**DONE**	P07 + deviation chain
CTL-4	Payment block until exceptions resolved	**DONE**	M10-R5
CTL-5	Every approval records actor, timestamp, comment, rule version, evidence shown	**DONE**	F-audit; `task.rule_set_version`

BP §25 AI agents (Table 25), autonomy levels (Table 26), answer contract

DB `agent_run` counts per agent are shown in brackets.

ID	Agent (default autonomy)	Status	Evidence / gap
AG-01	Demand Sensing (L1)	**DONE**	`svc/forecasting.py` (1 run; DC-331 bias recommendation accepted)
AG-02	MRP Exception (L2)	**DONE**	`svc/planning.py` (5)
AG-03	Guided Buying (L2): finds contract/catalogue matches, **fills account assignment**	**PARTIAL**	Registered, but **never invoked (0 runs)**. The search is plain service logic, and account assignment is not auto-filled.
AG-04	Material Dedup (L2)	**DONE**	`svc/mdm.py` (21)
AG-05	Spend Classification (L3 auto below threshold)	**DONE**	`svc/spend.py:182` guard (304)
AG-06	Sourcing Strategy (L1)	**DONE**	`svc/sourcing_strategy.py` (3)

ID	Agent (default autonomy)	Status	Evidence / gap
AG-07	Bid Analysis (L2)	**DONE**	`svc/sourcing.py` (4)
AG-08	Contract Review (L2)	**DONE**	`svc/contract.py` (17); deterministic, no LLM
AG-09	PO Validation (L3)	**DONE**	`svc/order.py:validate` (67)
AG-10	Touchless PO (L4 within policy)	**DONE**	`svc/order.py:343` guard (42)
AG-11	Expediting (L2)	**DONE**	`svc/order.py` (2)
AG-12	Invoice Exception (L2)	**DONE**	`svc/payables.py` (9)
AG-13	Supplier Risk (L1)	**DONE**	`svc/risk.py` (8)
AG-14	Savings Validation (L3 auto-compute, finance validates)	**DONE**	`svc/savings.py` (5)
AUT-L	Autonomy levels L0–L4 with meanings	**DONE**	`plat/agents.py:LEVELS`; L1 → RECOMMENDED, L2 → DRAFTED (`recommend`)
AC-BP	Answer contract fields, deep link never executes, low confidence stated	**DONE**	CP-1; confidence label "low" below 0.5
AG-QUAL	Phase-3 exit "6 agents in production"	**EXTERNAL**	Code is ready; production needs deployment and feedback volume (≥ 20 per agent before suspension logic applies)

BP §26 Savings methodology

ID	Item	Status	Evidence / gap
SAVM-1	Negotiated = (baseline − new) × forecast volume; never in budget	**DONE**	`svc/savings.py` ledger `counts=False`
SAVM-2	Avoidance = (requested/market − agreed) × volume	**DONE**	AVOIDANCE ledger (SV-0412, 0.318 Cr)
SAVM-3	Realised = (baseline − invoiced) × invoiced qty from posted invoices; counts after finance validation	**DONE**	ALG-9.5
SAVM-4	Baseline method fixed per initiative at creation	**DONE**	`savings_initiative.baseline_method`

BP §27 KPI framework (Table 28)

ID	Module KPIs	Status	Evidence / gap
KPI-M01..M13	55 KPIs across 13 modules	**PARTIAL**	About 26 are computed (AIX-01 `svc/insight.py:dashboard`, `/payables/insights`, `/supplier-performance`, `pic_slo_*`, forecasting MAPE, D9 accuracy). Missing per module: M02 backlog, conversion lead time, stock-outs, days of cover; M03 all; M04 channel %, free-text %, budget-blocked; M05 cycle time, competitive %; M06 onboarding cycle; M07 cycle time, obligation on-time; M08 PO cycle time; M09 all; M10 exception cycle, duplicates prevented; M11 realisation rate; M12 agent-assisted %, hours saved; M13 MTTR.

BP §28 Roadmap & MVP definition of success

ID	Criterion	Status	Evidence / gap
MVP-1	Touchless PO ≥ 30% for catalogue/contract lines in the pilot plant	**EXTERNAL**	Demo AXN shows 39%; it must be measured on the pilot
MVP-2	First-pass invoice match ≥ 80%	**EXTERNAL**	Demo AXN shows 72% (13/18); must be measured on the pilot
MVP-3	Zero duplicate ERP postings across 10,000 invoices in UAT	**EXTERNAL**	The mechanism is DONE (INT-12.2); never run at 10k volume or against a real ERP
MVP-4	Every PO line traceable to its origin in one click	**DONE**	Lineage tab on the PO page
RM-0..4	Phases 0–4 scope	**PARTIAL**	All phase scopes are built in code. The exits that depend on real ERP/connectors (Phase 1) and the workbook industry sheets (Phase 4) are open.

Project design decisions DD-01..11 (`docs/PIC_Delivery_Plan.md` §3; project-internal, not in the Blueprint)

ID	Decision	Status	Evidence / gap
DD-01	FastAPI modular monolith; no cross-module table access, enforced by import-linter	**PARTIAL**	Monolith is DONE; **no import-linter**, and cross-module table reads are common (TD-3a)
DD-02	PG 16 + RLS, SQLAlchemy, Alembic	**DONE**	AD-02
DD-03	Outbox to a Kafka-compatible broker, Avro registry	**NOT DONE**	EV-3
DD-04	Camunda 8 via pyzeebe	**NOT DONE**	Replaced by the own engine (AD-01)

ID	Decision	Status	Evidence / gap
DD-05	Keycloak broker (OIDC/SAML)	**NOT DONE**	Direct OIDC only; no SAML
DD-06	Postgres materialised views in MVP, ClickHouse from Phase 2	**NOT DONE**	No materialised views and no ClickHouse (AD-04)
DD-07	LLM only via LiteLLM	**NOT DONE**	AD-06 (dependency present, unused)
DD-08	Rule sets in Postgres, evaluated in Python	**DONE**	AD-07
DD-09	Supplier portal as a second Next.js app	**NOT DONE**	FE-4 (same app)
DD-10	Twin as a separate bounded context PIC reads from	**PARTIAL**	INT-TWIN
DD-11	GCP asia-south1 + asia-south2 DR, Terraform	**EXTERNAL**	Terraform ready (no DR stack); current hosting decision is Vercel + Neon Singapore

Part 3 — Datasets D1–D11

Source: `backend/app/pic/seed/datasets/*.py` packs, `pic.dataset_load` in pic_neon_build, and `tests/test_datasets_*.py`.

#	Dataset	Status	Evidence (pack · DB · tests) / gap
D1	Branch cases for every unexercised gateway (cases 9–34)	**DONE**	`branches.py` (dataset `axn-branch-cases`). All 84 steps executed and 13/13 gateways taken both ways. 25 tests in `test_datasets_branches.py`. Data gaps: no CANCELLED sourcing event, no LOST savings, no PREFERRED supplier and no RENEWED contract remain in the final data (some are tested in the suite only).
D1b	Operations module cases (amendments, auction, EDI, e-way bill, 4-way, reconciliation, payment files, MDM-04, punch-out, ERP import)	**DONE**	`module_cases.py` (`axn-module-cases`: PO-77425, RFQ-2233, IB-101, REC-102/103, PF-1001/1002, REQ-2330); 8 tests. The e-sign leg records "ESIGN_NOT_CONFIGURED".
D2	Purchase types: expense, capital asset, consignment, subcontracting	**DONE**	`purchase_types.py` (`axn-purchase-types`); 9 tests
D3	Construction tenant CON: WBS C-2207, SC-118, SES 35%, 5% retention, VO	**DONE**	`industry_construction.py`. CON: 6 suppliers, 2 POs, 5 invoices, 43 spend lines, 1 contract. 6 tests. Advance recovery and bank guarantee are not modelled (IND-CON).
D4	IT services ITS: SOW-4410, RC-22, 320 h SES, licence renewal	**DONE**	`industry_it.py`. ITS: 6 suppliers, 2 POs, 2 invoices, 40 spend lines, 2 contracts. 5 tests.
D5	Logistics LOG: lane CHE→PNQ ₹54/km + fuel index, POD, freight audit	**DONE**	`industry_logistics.py`. LOG: 16 POs, 16 invoices. 7 tests. Fuel index is demo data.
D6	Healthcare HLT: KN-77 consignment, lot/expiry, usage PO, recall	**DONE**	`industry_healthcare.py`. HLT: 8 POs, 3 invoices plus ERS settlements. 7 tests.
D7	General GEN: laptop punch-out ₹78,500, single approval	**DONE**	`industry_general.py`. GEN: 9 POs, 13 invoices. 5 tests. Punch-out uses the simulator.
D8	Volume: 50k materials, 5 M invoices/yr, 50 M spend lines, 10k users	**PARTIAL**	Generator `tools/volume_gen.py` + `tools/perf_mrp.py`; 3 tests (`test_volume_gen.py`). It is not loaded into pic_neon_build, by design. **Only the 50k-material MRP run was executed (681–1,615 s, FAIL). 5 M invoices and 50 M spend lines were never generated or measured.** The k6 10k-user test was never run. Needs a production-like environment.
D9	AI evaluation sets (2,654 labelled spend lines, 45 forecast points, 48 duplicate pairs, 40 Copilot questions)	**DONE**	`ai_eval.py` + `tools/ai_eval.py` (`ai-evaluation-sets`); 7 tests. Gates: classification 0.9936 PASS, MAPE h1 0.037 PASS, Copilot 1.0 PASS, dedup recall 1.0 PASS. **autoApplyPrecision 0.857 < 0.95 FAIL; dedupPrecision 0.83 < 0.90 FAIL.** The gate targets are "proposed" and need your sign-off.

#	Dataset	Status	Evidence (pack · DB · tests) / gap
D10	Real tenant masters from your ERP (vendors with bank/tax, materials, contracts, open POs, open AP, 24–36 months history)	**EXTERNAL**	Only the legacy prototype DB was migrated: GMC (5 suppliers, 0 materials, 5 POs, 2 spend lines) and DBY (1 supplier, 9 materials, 6 POs, 13 spend lines). AXN history (2,568 lines, 20 ERP vendors) is **generated**. Loaders for open contracts, open POs and open AP are not built (MIG-2/3/5).
D11	Reference data: FX, HSN/SAC + GST, holiday calendars, UNSPSC/taxonomy	**PARTIAL**	`reference_data.py` (`reference-data` loaded for all 8 tenants); 9 tests. HSN-GST: 370 CBIC entries. TAXONOMY-UNSPSC: 56 mappings. Holidays: 23–46 per plant, **but the migrated GMC and DBY P100 plants have none**. **FX is a single day (28 Oct 2026; 4 rates per tenant), with no history.** 2027 state holiday lists are incomplete. UNSPSC is a mapping only, not the licensed code set.

Part 4 — Summary counts

Counted from the Status column of every row above.

Document	DONE	PARTIAL	NOT DONE	EXTERNAL	N/A	Total
Technical Document (Part 1)	82 (46%)	68 (38%)	17 (10%)	10 (6%)	0	177
Blueprint (Part 2, incl. 11 project DD rows)	199 (66%)	83 (28%)	12 (4%)	7 (2%)	0	301
Datasets (Part 3)	9	2	0	1	0	12
All	**290**	**153**	**29**	**18**	**0**	**490**

Blueprint without the project DD rows: 197 DONE, 81 PARTIAL, 6 NOT DONE, 6 EXTERNAL (290 rows).

What the numbers say:

- The **business logic of the Blueprint is largely built and tested**. That covers flows F1–F9, state machines, DMN tables, SoD, purchase types, the five industry tenants, the agents and the signature case.
- The **Technical Document's platform choices and NFRs are where the gaps are**: Camunda, Kafka, ClickHouse, the LiteLLM gateway, GraphQL, a separate portal app, bpmn-js, Playwright/Pact, measured performance and accessibility.
- **Real integrations are also open**: a real SAP connector, ERP inbound sync and migration loaders.

Part 5 — What you must supply or decide

De-duplicated. The last column shows the rows each item unblocks.

5.1 Decisions (architecture, product, legal, tax)

#	Decision	Unblocks
DEC-1	**Accept or reject the architecture deviations**: FastAPI monolith instead of NestJS services; Postgres workflow engine instead of Camunda 8; in-process outbox instead of Kafka/Avro; Postgres spend cube instead of ClickHouse; REST only (no GraphQL); portal in the same app. Accepting them closes about 15 PARTIAL/NOT DONE rows. Rejecting them is L-size work per item.	TD-1.1a, TD-2.1b–e, AD-01, AD-03, AD-04, TD-3a, EV-3, API-1, FE-4, TST-C, DD-01/03/04/06/09
DEC-2	**Hosting and residency**: stay on Vercel (Pro or Hobby) + Neon Singapore, or move to GCP asia-south1 (India residency, DD-11). Also decide on tenant-level residency.	TD-2.1h, NFR-RES, NFR-AV, NFR-DR, SEC-PRIV, DD-11
DEC-3	**Neon plan** (PITR ≥ 7 days, exports) to meet RPO 15 min / RTO 4 h; DR region	NFR-DR
DEC-4	**Identity**: corporate IdP via OIDC, and whether **SAML** is required; switch MFA `enforce=true` per tenant; mandatory MFA for supplier users	SEC-ID, NFR-SEC
DEC-5	**LLM**: provide the LiteLLM gateway (URL, key, model list), or confirm the deterministic Copilot and playbook redline are acceptable	AD-06, CP-3, ML-8, DD-07
DEC-6	**Rule values the spec leaves open**: touchless value band (default ₹25 L); DMN-01 band edges at exactly ₹2 L / ₹10 L; PO-deviation rule (contract tolerance vs +2%, and last price +5%); escalation (add manager vs reassign); AI-eval gate targets (auto-apply precision ≥ 0.95, dedup precision ≥ 0.90)	DMN-01, DMN-02, APR-8, F-esc, ML-1, ML-3, TST-AI
DEC-7	**Legal**: retention periods per data category (10-year audit vs the current 8-year default), lawful-basis wording	NFR-AUD, SEC-PRIV
DEC-8	**Tax team**: GST on goods transport (SAC 996791), implant rate, ERS self-billing under GST, job-work SAC 9988; GST rate on material requests	PT-7, IND-HLT, IND-LOG, M10-C2
DEC-9	**Product scope**: multi-item/lot sourcing events (currently one item per event); which industry template and purchase types the pilot uses first	ALG-9.3, M05-C1/C2, IND-*
DEC-10	**ZAP gating**: triage the baseline, then make ZAP blocking	SEC-APP, TST-SEC

5.2 Systems, credentials and access

#	Supply	Unblocks
SUP-1	**SAP S/4HANA sandbox** (OData service URLs, technical user; IDoc if needed), or the target ERP of your pilot (Oracle/D365/Tally gateway/Odoo credentials)	INT-SAP, INT-ORA, INT-D365, INT-TALLY, INT-ODOO, INT-IN, M03-C7, M10-C6, P13, PH-1, MVP-3
SUP-2	**GST IRP/GSP** credentials (`PIC_IRP_*`)	INT-GST, M10-C1
SUP-3	**E-signature provider** (`PIC_ESIGN_PROVIDER/URL/API_KEY/WEBHOOK_SECRET`)	M07-C4, F4
SUP-4	**Bank host-to-host/SFTP** and the real debtor account/IFSC (PAYMENT-FILE rule)	M10-C7
SUP-5	**EDI gateway** (AS2/SFTP, `PIC_EDI_GATEWAY_URL/TOKEN`) and supplier EDI IDs; **punch-out supplier accounts**	INT-EDI, INT-PUNCH, M08-C5, D7
SUP-6	**Sanctions feed** licence (`PIC_SANCTIONS_*`), plus financial-risk and news providers if wanted	INT-RISK, M06-C6
SUP-7	**Supplier Capability Digital Twin** API endpoint and credentials (capability, free capacity, trust, late-delivery)	INT-TWIN, ML-5, M06-D, DD-10
SUP-8	**SMTP** (`PIC_SMTP_*`) for PO e-mails, invitations and notifications	M08-C5, F8
SUP-9	**OCR/IDP engine** (`PIC_OCR_URL/TOKEN`)	M10-C1
SUP-10	**Hosting access**: Vercel team/token and plan, or a GCP project; GitHub secrets (`NEON_DATABASE_URL` , `PIC_JWT_SECRET` , `PIC_FIELD_KEY` , `PIC_METRICS_TOKEN` , `PIC_SEED_PASSWORD`); rotated Neon password; Neon URL (still not provided); KMS key if on GCP	TD-2.1h, SEC-DP, TD-4b
SUP-11	**OTel collector / log sink** (Cloud Logging or Grafana)	OBS-1, DOD-6
SUP-12	**A staging environment** sized like production, for k6, cube, MRP and 10k-user runs	NFR-P95, NFR-PR, NFR-CUBE, NFR-MRP, NFR-SCALE, D8, TST-PERF

5.3 Data and documents

#	Supply	Unblocks
DAT-1	**ERP extract** (24–36 months PO + invoice history, vendors with bank/tax IDs, materials, open contracts, open POs, open AP, GL totals for tie-out)	D10, MIG-1..5, SIG-11, MVP-1/2
DAT-2	**FX history** (FBIL archive) and the currency of legacy amounts	D11, TD-1.1c
DAT-3	**Reference data**: 2027 holiday lists for the remaining states; holidays for the migrated P100 plants; real IFSC codes; a real diesel/fuel index feed; licensed UNSPSC (if class level is needed); your spend taxonomy	D11, IND-LOG, M01-C2
DAT-4	**Test-Case Workbook AHG/PRC/PIC/2026-10/03** (E2E-01..n, industry test sheets)	TST-E2E, PH-4, DOD-1
DAT-5	**Business data**: FY 2027-28 budgets, NPPA ceiling prices for implants	D1/D2 budgets, IND-HLT

Part 6 — Top gaps you can build without the customer (ranked)

Size: S ≈ ≤ 1 day, M ≈ 2–5 days, L ≈ 1–3 weeks.

Rank	Gap	Rows	Size
1	**Apply Idempotency-Key to all 112 unprotected POST routes**; add the 24 h expiry/purge; add ETag/If-Match on all mutable documents (requisition, contract, invoice, sourcing event, supplier)	API-2, API-3, DOD-4	M
2	**Enforce ABAC on commands** (company code, plant, category, value band: wire `check_attr` / `check_value_limit` into the services); add the SoD mitigating-approval path; add the SoD check inside the agent guard and call the guard for every acting agent	SEC-AZ, SEC-SOD, AGT-2, SOD-AGT	M
3	**MRP to < 600 s at 50k materials** (drop the per-order link rows or batch them, summary response, COPY for planned orders); add weekly buckets by MRP type	NFR-MRP, ALG-9.1, M02-NFR	M
4	**Spend at scale**: rewrite SPD-01 `overview` as SQL aggregates; add materialised views (DD-06) with refresh; add period/filters, price-variance and invoice/PO-count measures to the cube; run `volume_gen --measure` at 50 M rows	AD-04, NFR-CUBE, M01-C1/C5/D	M–L
5	**Scheduled jobs**: nightly reconciliation per company code, payment-due timer, monthly scorecard recompute, retention purge, idempotency purge, scheduled MRP. Requires Vercel Pro or the GCP worker for minute ticks.	INT-12.2-5, F7, M06-R4, SEC-PRIV, NFR-AUD	S–M
6	**Test layers**: Playwright E2E for the signature case and the 5 industry packs; a coverage gate (85% domain logic); event JSON-schema contract tests; axe accessibility checks in CI	TST-E2E, TST-U, TST-C, NFR-A11Y, UX-6, DOD-5	M–L
7	**KPI completion**: compute the ~29 missing module KPIs (M02 backlog/days of cover, all of M03 and M09, cycle times, realisation rate, agent-assisted %, MTTR) and the template KPIs (PPV, cost to complete, cost per tonne-km, …)	KPI-M01..M13, IND-KPI	M
8	**Front-end gaps**: SPD-08 pipeline tracker, lifecycle strip, generic Approvals/Documents tabs, a bpmn-js process viewer, saved views and bulk actions; **fix the hard-coded "Manufacturing template" header** (S)	FE-1/2/3/5, UX-1/3, M01-S	M
9	**Industry and purchase-type controls that are only flags or comments**: clinical sign-off as an award approval step, advance recovery, performance bank guarantee tracking, accessorials/detention, ERS for scheduling agreements, Kanban/reorder point, licence utilisation, P-card; price-history check ("last price + 5%") on PO validation; risk constraints in award optimisation	IND-*, APR-8, M08-C3, ALG-9.3	M–L
10	**Master data and the org model**: legal-entity and purchasing-org/group masters (remove `purch_org="1000"` and `currency="INR"` hard-codes), configurable mandatory fields, material change requests with impact analysis, sales-order account assignment, real FX conversion with the rate and date stored on amounts	TD-4a, TD-1.1c/f, G-1, M03-C5, M04-C4	M–L
11	Migration loaders (contracts, open POs, open AP, spend history with GL tie-out), ready for when the ERP extract arrives	MIG-1..5	M
12	Event consumers per the catalogue (ai-svc consumers that feed agents, supplier.status.changed → open-PO/RFQ flagging); a published event schema document	EV-4, DOD-2	S–M

Rank	Gap	Rows	Size
13	Field-encryption key versioning and a re-encryption job; breached-password check; a written threat model	SEC-DP, SEC-APP	M
14	Data fixes: seed enough posted invoices so SV-0412 realised ≈ 0.38 Cr and first-pass ≈ 86%; add holidays for the GMC/DBY P100 plants	M11-D, SIG-10/11, D11	S